

Tutorial 4

Statistical Computation and Analysis
Spring 2026

Tutorial Outline

- Short revision on lecture 4
- Lecture 5 models

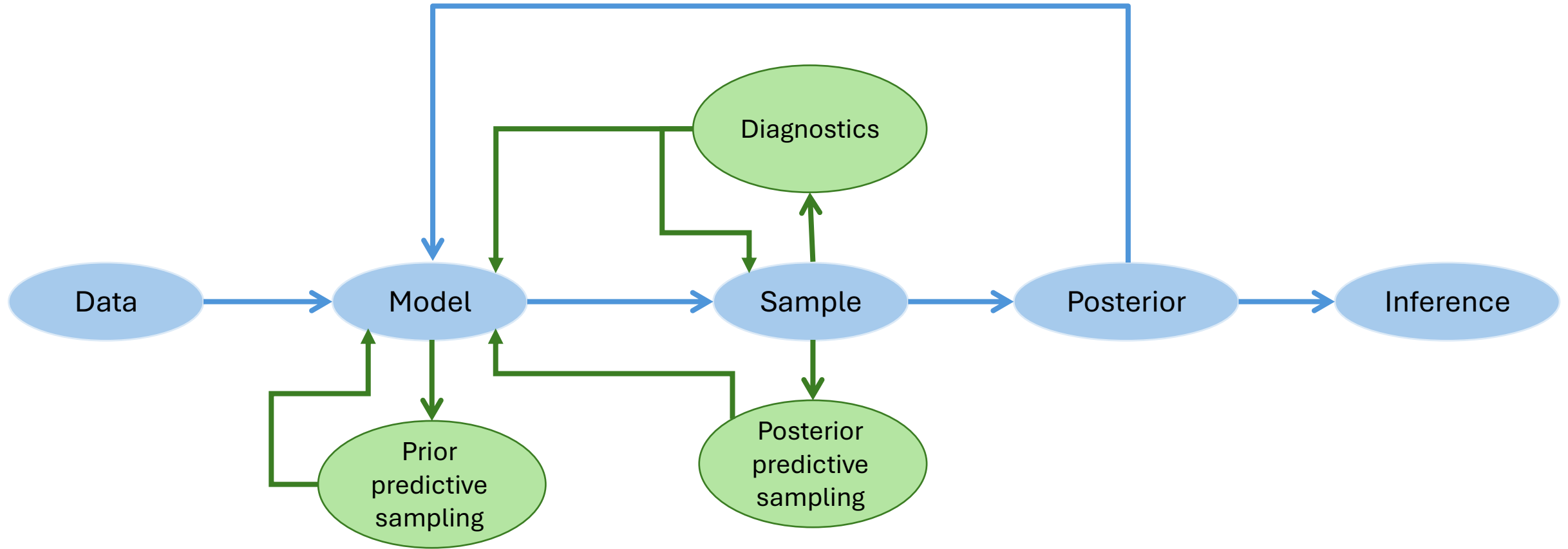
Revision

- Bayesian workflow
- Sampling diagnostics
- Divergences
- Validation
 - Prior predictive plots
 - Posterior predictive plots

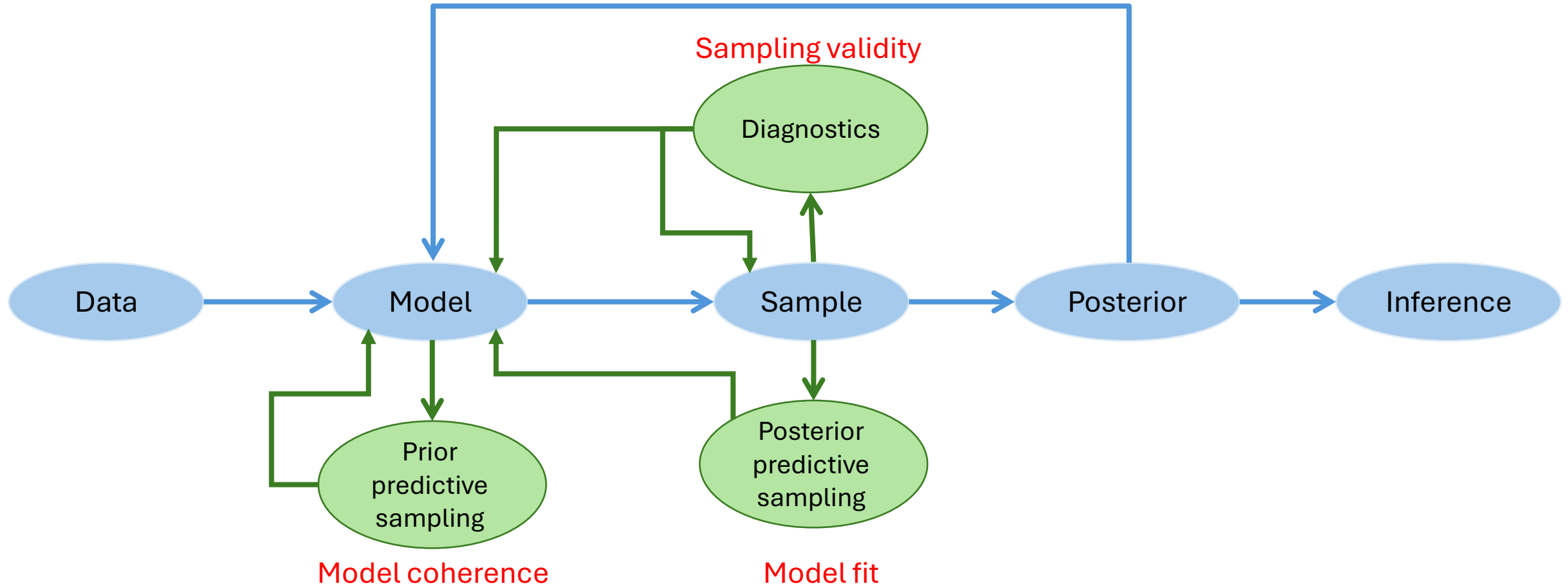
Bayesian workflow



Bayesian workflow

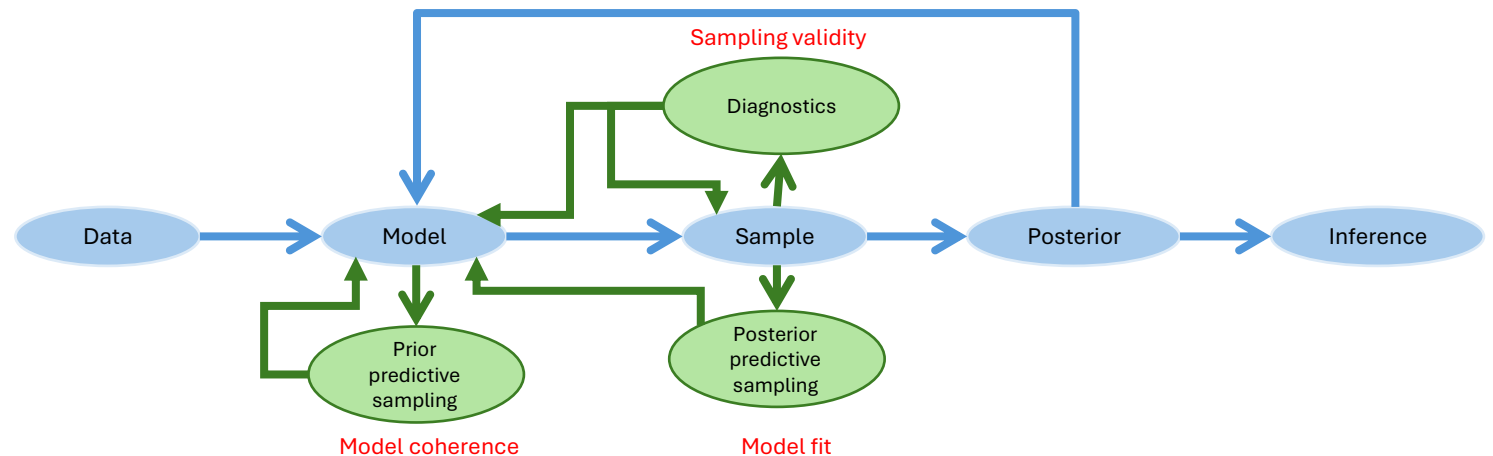


Bayesian workflow



Bayesian workflow

- **Prior predictive sampling** - simulating parameters from the prior and data from the likelihood before fitting the model.
- **Sampling validity** - whether the algorithm actually sampled from the posterior distribution.
- **Model fit** - whether the fitted model can generate data resembling the observed data.
- **Posterior predictive sampling** - simulating new data from parameter values drawn from the posterior.
- **Iterative workflow** - the process of revising data procedures, priors, models, parameterizations, or sampler settings when checks fail.



Sampling diagnostics

- Posterior Summary

```
az.summary(  
    idata_pooled,  
    var_names=["theta"]  
)
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.075	0.001	0.072	0.076	0.0	0.0	1297.0	1308.0	1.0

Sampling diagnostics

- Posterior Summary

kind='stats'

```
# posterior summary
az.summary(
    idata_pooled,
    kind="stats",
    var_names=["theta"]
)
```

	mean	sd	hdi_3%	hdi_97%
theta	0.075	0.001	0.072	0.076

kind='diagnostics'

```
# posterior summary
az.summary(
    idata_pooled,
    kind="diagnostics",
    var_names=["theta"]
)
```

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1297.0	1308.0	1.0

Sampling diagnostics

- Trace plots & divergences
- $\hat{R}$
- ESS
- MCSE

Sampling diagnostics

- $\hat{R}$, ESS and MCSE:
 - Diagnostic metrics used to evaluate whether samples generated by Markov Chain Monte Carlo (MCMC) methods are reliable and have converged properly.

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1297.0	1308.0	1.0

Sampling diagnostics

- **Effective Sample Size (ESS)** - the number of independent draws with roughly equivalent information content.
- MCMC methods use the current position to generate the next position, the samples they produce are correlated. Because of this autocorrelation, the actual number of "useful" or independent draws is lower than the total number of samples generated.
- You can check the **bulk-ESS** to see how well the center of the distribution was resolved, or the **tail-ESS** if you are interested in reporting extreme quantiles or rare events.

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1297.0	1308.0	1.0

Sampling diagnostics

- $\hat{R}$ - a convergence diagnostic comparing between-chain variation to within-chain variation.
- Ideally we want $\hat{R} = 1$.

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1297.0	1308.0	1.0

Sampling diagnostics

- **Monte Carlo Standard Error** - Uncertainty in posterior summaries caused by finite sampling.
- Should be less than 1–2%.

$$\text{MCSE} \approx \frac{\text{sd}(\theta)}{\sqrt{\text{ESS}}}$$

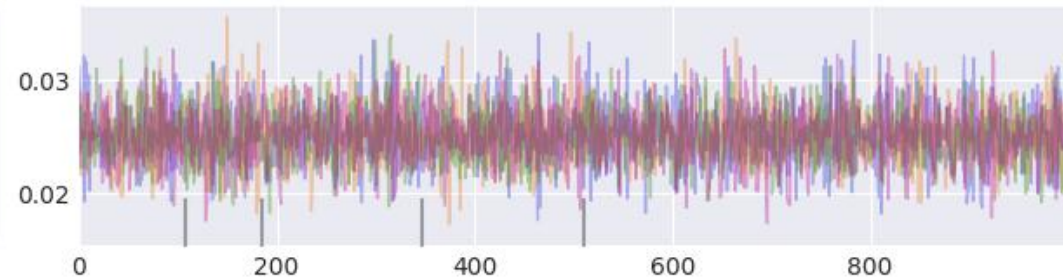
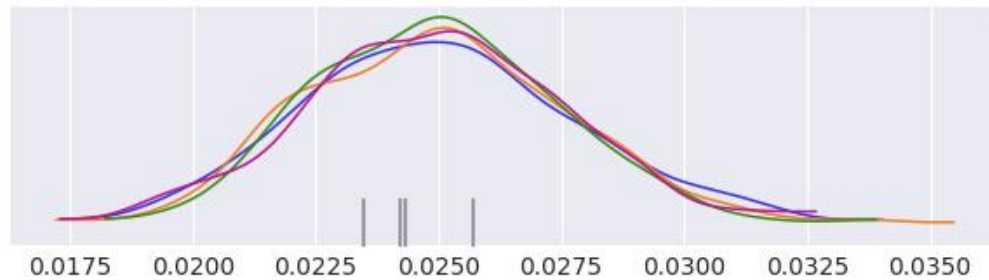
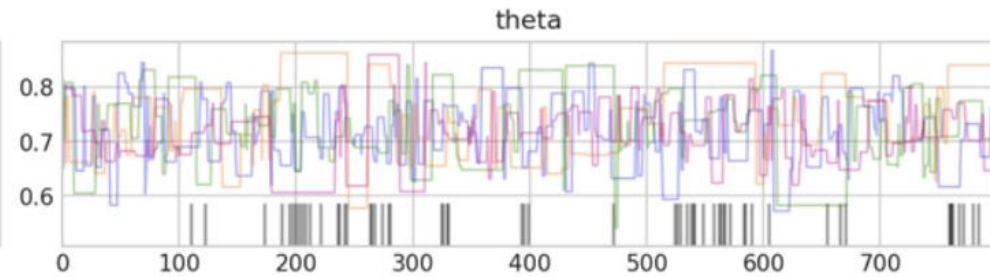
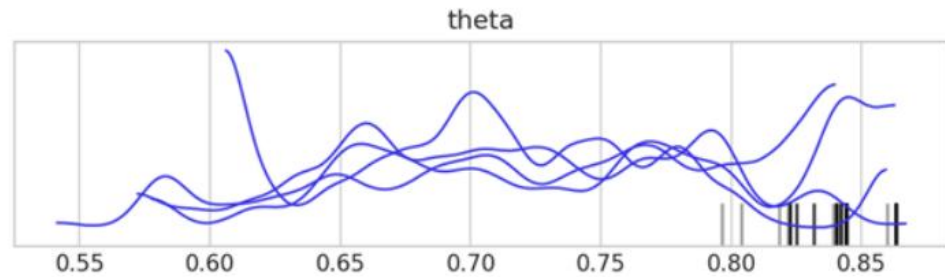
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1297.0	1308.0	1.0

Divergences

- Diagnostic warning exclusive to the NUTS (No-U-Turn Sampler) method.
- Unlike the other that evaluate the properties of the generated samples, divergences are based on the inner workings of the sampling algorithm itself.
- How to fix:
 - **Increase the target_accept parameter:** force the sampler to take smaller, more careful steps and resolve the divergences. Default is 0.8, If you have to push this value extremely high (e.g., 0.999 it might not be enough.
 - **Reparameterize the model:** rewriting the model in a mathematically equivalent way that creates a friendlier geometry for the sampler.

Divergences

- Trace plots with divergences examples



Validation

Reminder

- Prior predictive sampling shows model coherence
- Diagnostics shows sampling validity
- Posterior predictive sampling shows model fit

Validation

- **A prior predictive plot** shows data simulated from the model before seeing the observed data.
- If my priors and likelihood are true, what kind of data could this model generate?
- Usually we want the prior predictive samples to be wider than the observed data.

Validation

- **A posterior predictive plot** shows data simulated from the model after fitting the model.
- After learning from the observed data, can the model generate data that look like the real data?
- Usually we want to see if the fitted model reproduce the important patterns in the observed data

Lecture 5

- Pooled & unpooled models
- Hierarchical model
- Multi-level models

Beta Distribution – Clarification

Up until now we used alpha and beta:

```
pz.Beta(alpha=3, beta=25).plot_interactive()
```

alpha (0, inf) 3.00

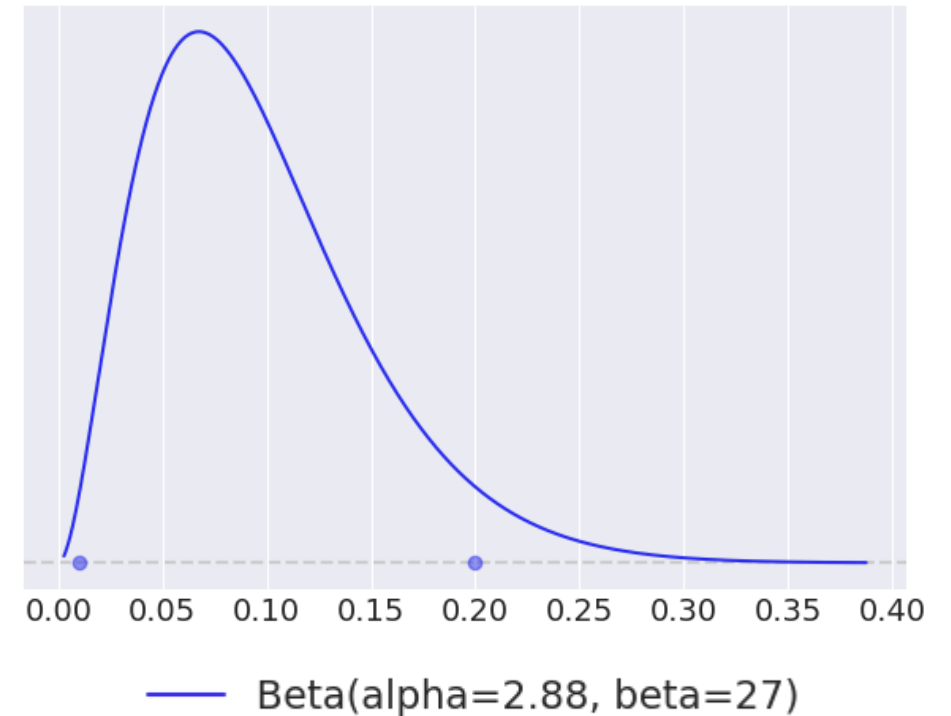
beta (0, inf) 25.00

Another function worth knowing:

```
biotech_rate_prior, ax = pz.maxent(  
    pz.Beta(),  
    lower=0.01,  
    upper=0.20,  
    mass=0.95,  
    plot=True  
)
```

```
omega_alpha = float(biotech_rate_prior.alpha)
```

```
omega_beta = float(biotech_rate_prior.beta)
```



Beta Distribution

- The same Beta distribution can be written using different parameterizations
- Alpha and beta:
 - α = success-like prior information
 β = failure-like prior information
 - Mean centered around $\frac{\alpha}{\alpha+\beta}$
 - `pm.Beta("theta", alpha=2, beta=18)`

Beta Distribution

- We can also use mu and nu:
 - mu = mean of the Beta distribution
nu = concentration around that mean
 - $\mu = \frac{\alpha}{\alpha + \beta}$
 $\nu = \alpha + \beta$
- `pm.Beta("theta", mu=0.10, nu=20)`

Beta Distribution

- Or use mu with sigma:
 - sigma: standard deviation / width
 - `pm.Beta("theta", mu=0.10, sigma=0.05)`

$$\text{Mean: } \mu = \frac{\alpha}{\alpha + \beta}$$

$$\text{Variance: } \sigma^2 = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} \quad \text{or}$$

$$\text{Standard deviation: } \sigma = \sqrt{\sigma^2}$$

$$\alpha = \mu\nu$$

$$\beta = (1 - \mu)\nu$$

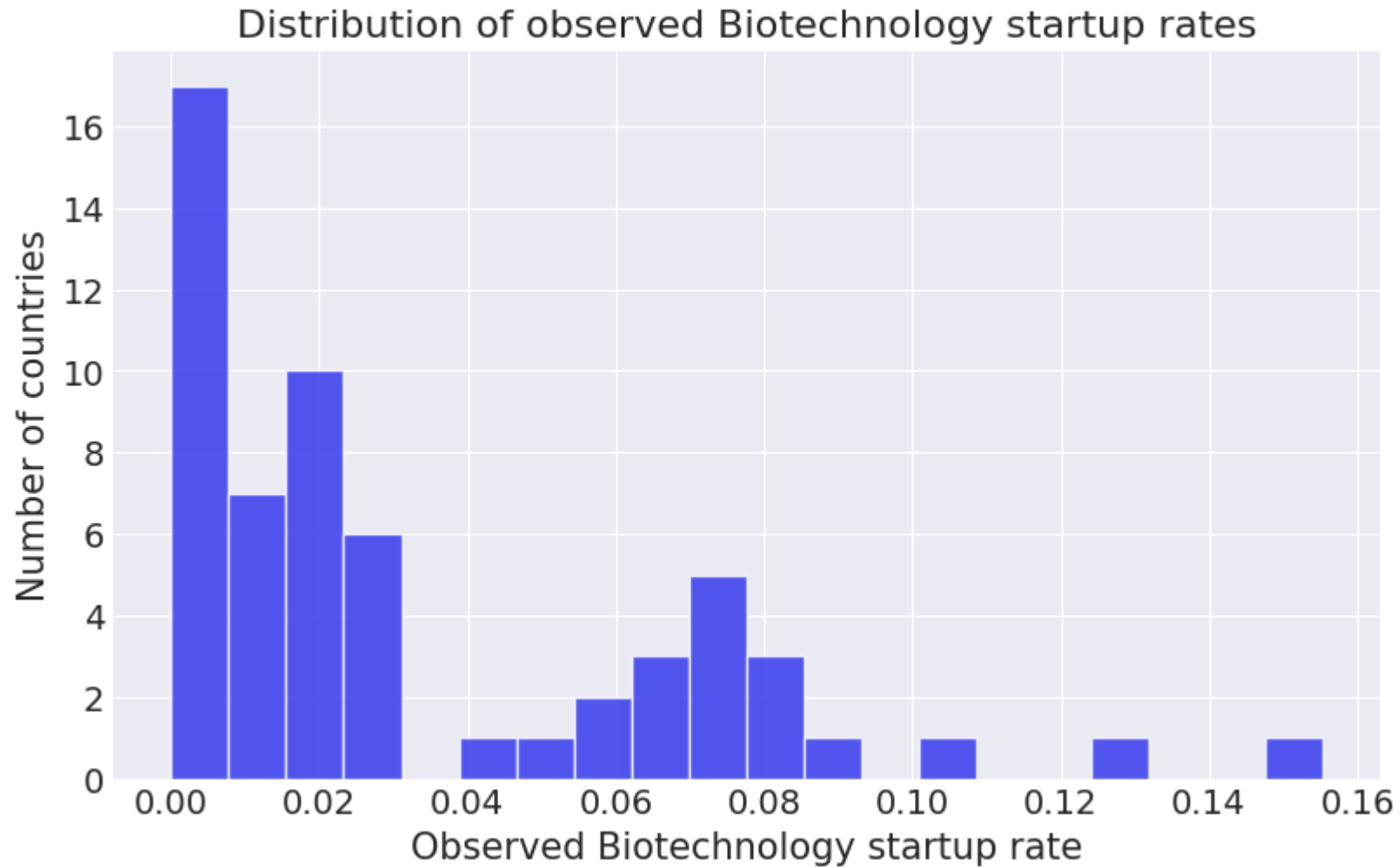
$$\sigma = \sqrt{\frac{\mu(1 - \mu)}{\nu + 1}}$$

Today's Dataset

- Startup dataset

	country_code	n_startups	n_biotech	observed_biotech_rate
0	USA	37601	3342	0.088881
1	GBR	3688	235	0.063720
2	CAN	1925	147	0.076364
3	IND	1596	22	0.013784
4	CHN	1544	47	0.030440

Today's Dataset



Our current workflow (code wise)

1. Load & describe at the data
2. Choose a prior (and hyperpriors)
3. Write the model + plot it as a graph
4. Prior predictive check (sample & plot)
5. Sample & diagnostics (posterior summary, trace plot)
6. Posterior predictive check (sample & plot)
7. Posterior (plot & HDI)

Go back and change at any point if needed

Pooled & Unpooled Models

Pooled Models

- A model that ignores group structure and estimates a single parameter from all observations together.
- The “baseline model, what we’ve done until now.

```
# Pooled Beta-Binomial model
coords = {"country": country_names}

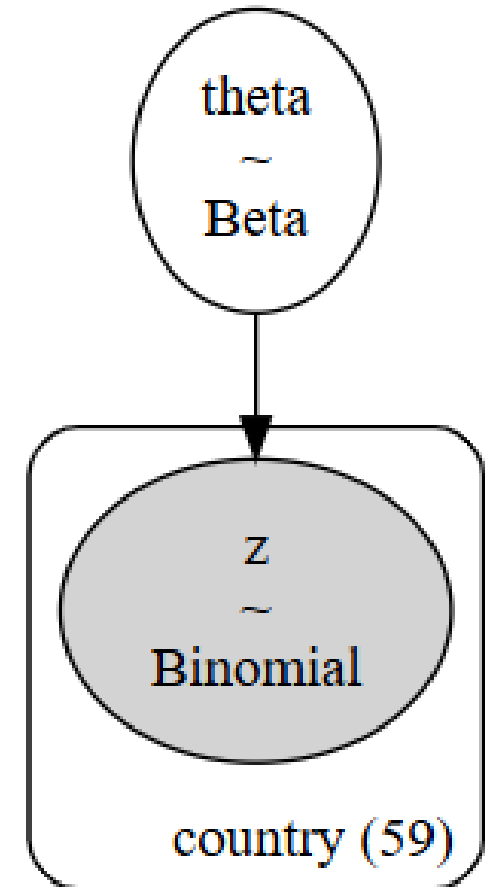
with pm.Model(coords=coords) as pooled_model:
    # prior
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta)
    # likelihood
    z = pm.Binomial("z", n=n_startups, p=theta, observed=n_biotech, dims="country")
```

Pooled Models

- As a graph:

```
# Pooled Beta-Binomial model
coords = {"country": country_names}

with pm.Model(coords=coords) as pooled_model:
    # prior
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta)
    # likelihood
    z = pm.Binomial("z", n=n_startups, p=theta, observed=n_biotech, dims="country")
```



Pooled Models

- Prior Predictive Check

1. Sample:

```
with pooled_model:  
    idata_pooled_prior = pm.sample_prior_predictive(  
        samples=1000,  
        random_seed=random_seed  
    )
```

Pooled Models

- Prior Predictive Check

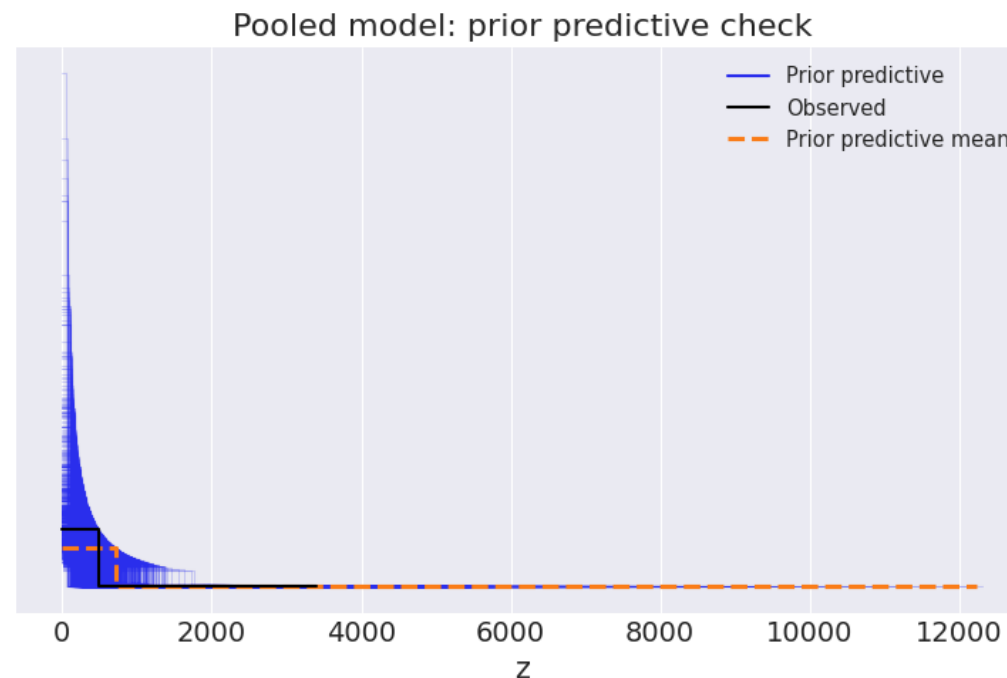
1. Sample

2. Plot:

1. `az.plot_ppc()` – for “z”
(number of startups in biotech).

- * Mainly used for count data in the discrete models

```
az.plot_ppc(  
    idata_pooled_prior,  
    group="prior",  
    observed=True  
)
```



Pooled Models

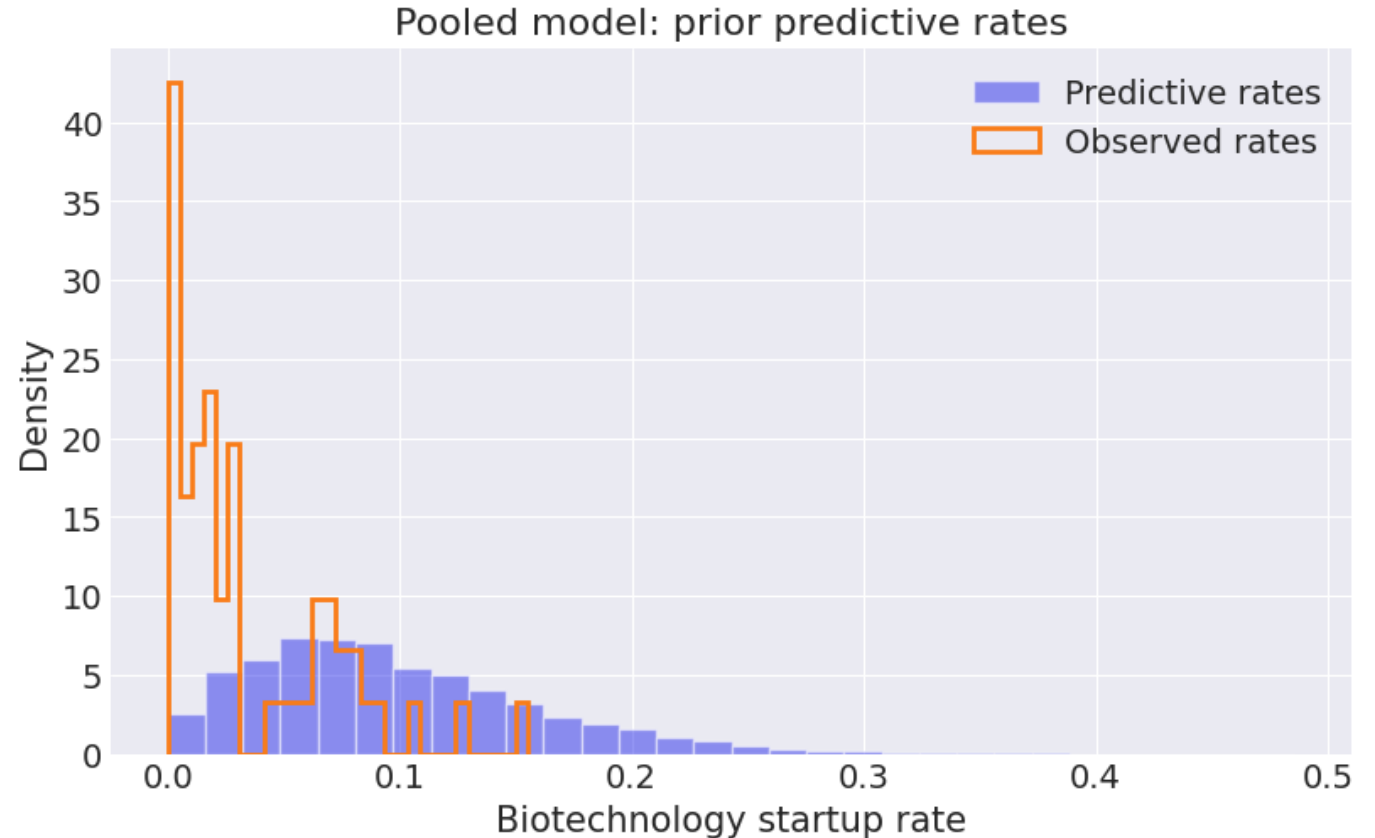
- Prior Predictive Check

1. Sample

2. Plot:

2. Create a function that extracts the idata, converts the counts to rates and plots it using `plt.hist()`

*Note – I created a “helper function” to plot both the prior & posterior predictive.



Pooled Models

- Sample

```
with pooled_model:
    idata_pooled = pm.sample(
        draws=1000,
        chains=4,
        random_seed=random_seed)
idata_pooled
```

Progress	Draw	Divergences	Step size	Grad evals	Speed	Elapsed	Remaining
	2000	0	2.084	1	840.66 draws/s	0:00:02	0:00:00
	2000	0	1.569	3	611.36 draws/s	0:00:03	0:00:00
	2000	0	1.442	3	476.43 draws/s	0:00:04	0:00:00
	2000	0	1.529	3	313.94 draws/s	0:00:06	0:00:00

arviz.InferenceData

- ▶ posterior
- ▶ sample_stats
- ▶ observed_data

Pooled Models

- Posterior summary

```
az.summary(  
    idata_pooled,  
    kind="diagnostics",  
    var_names=["theta"]  
)
```

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.0	0.0	1542.0	2359.0	1.0

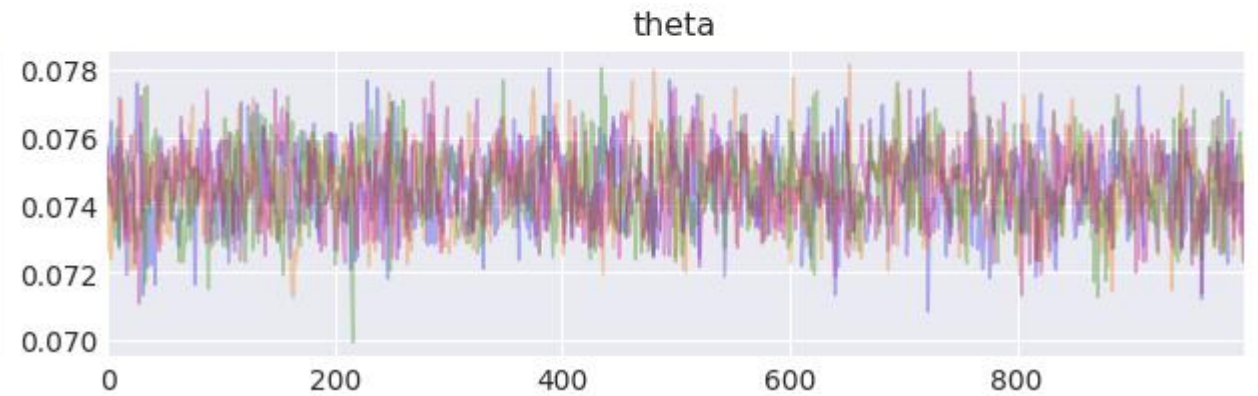
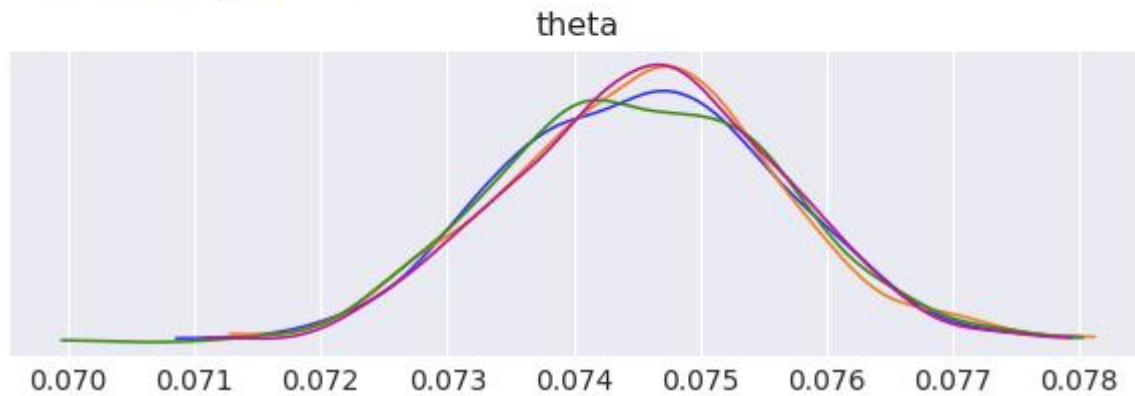
Pooled Models

- Diagnostics

```
az.plot_trace(  
    idata_pooled,  
    var_names=["theta"],  
    compact=False  
)
```

```
pooled_divergences = idata_pooled.sample_stats["diverging"].sum().values  
print(f"Number of divergences: {pooled_divergences}")
```

Number of divergences: 0



Pooled Models

- Posterior Predictive Check

1. Sample the posterior:

```
with pooled_model:  
    idata_pooled = pm.sample_posterior_predictive(  
        idata_pooled,  
        var_names=["z"],  
        random_seed=random_seed,  
        extend_inferencedata=True  
    )
```

Pooled Models

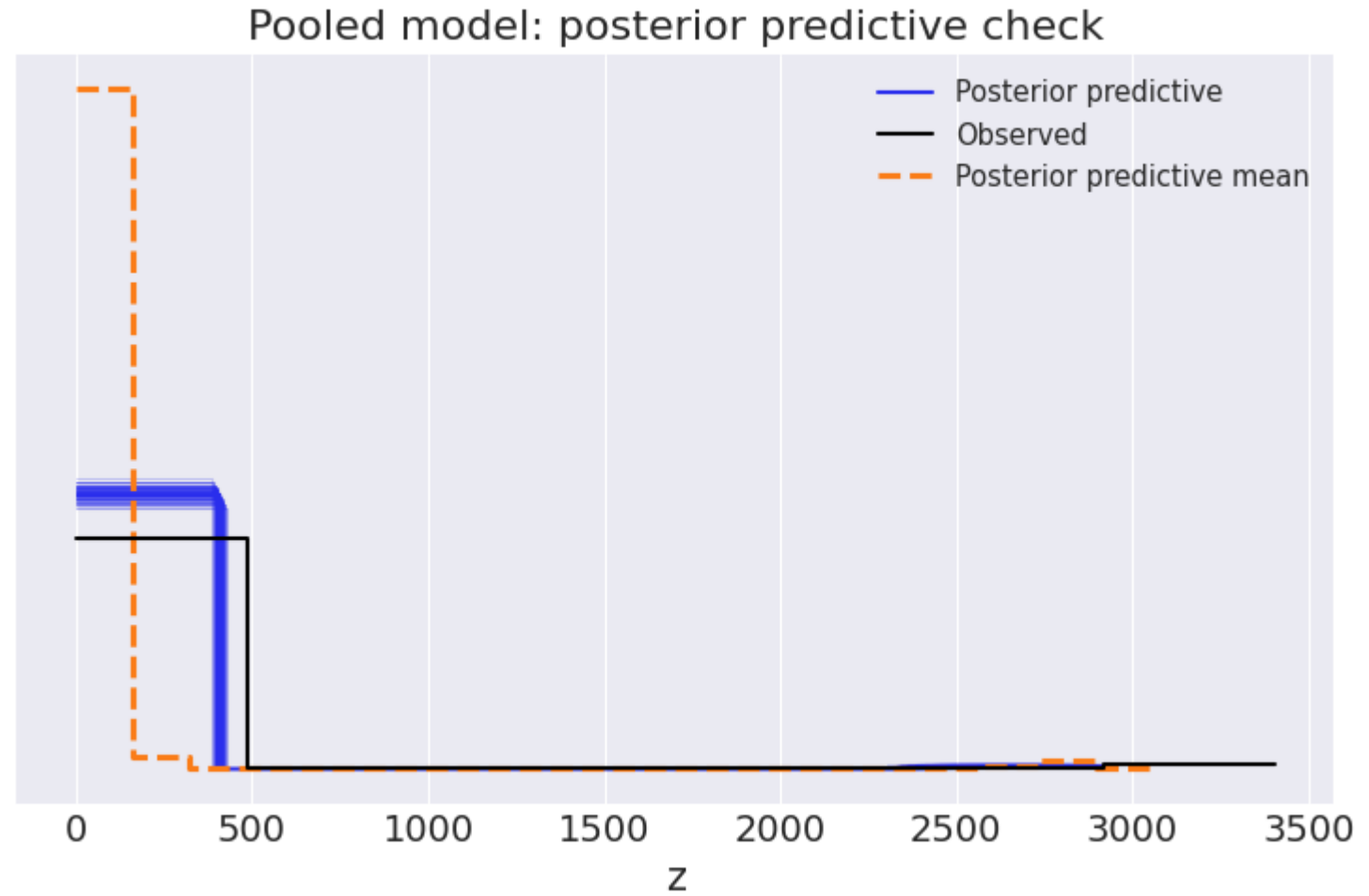
- Posterior Predictive Check

1. Sample the posterior

2. Plot:

1. Using `az.plot_ppc()`

```
az.plot_ppc(  
    idata_pooled,  
    group="posterior",  
    observed=True,  
    num_pp_samples=100  
)
```



Pooled Models

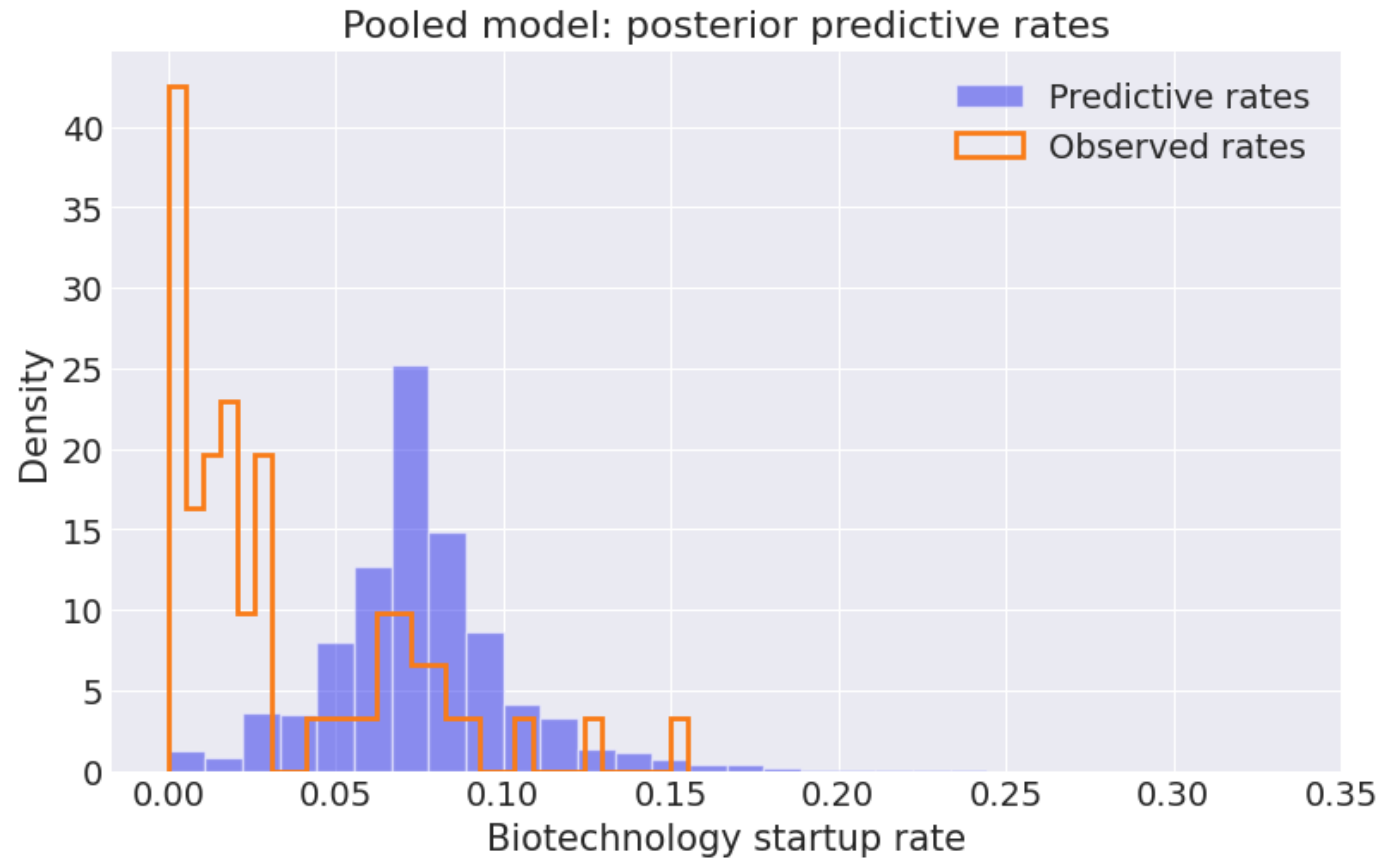
- Posterior Predictive Check

1. Sample the posterior

2. Plot:

2. Using our helper function:

```
plot_predictive_rates(  
    idata=idata_pooled,  
    group_name="posterior_predictive",  
    n_trials=n_startups,  
    observed_rates=observed_rate,  
    title="Pooled model: posterior predictive rates"  
)
```

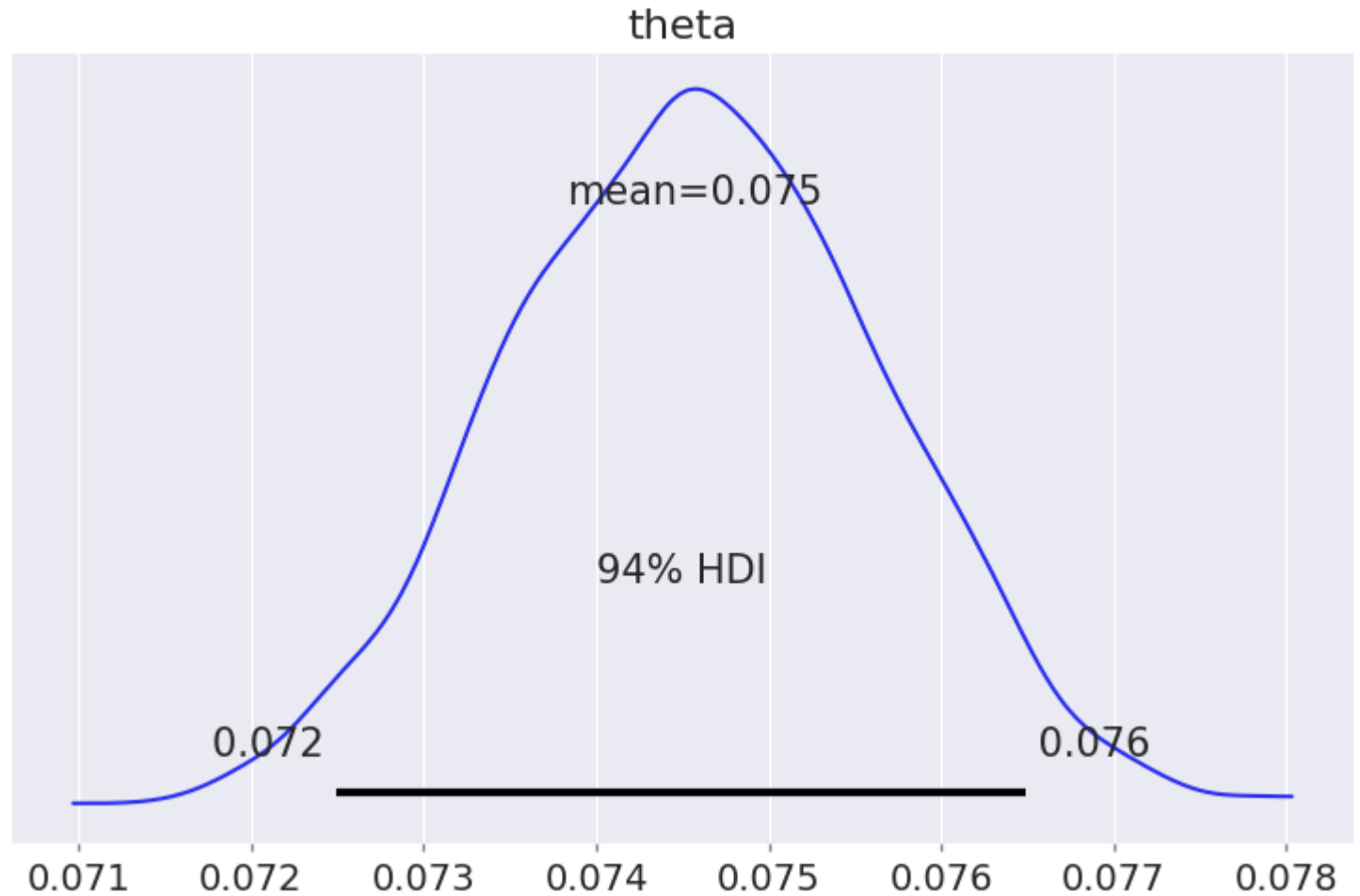


Pooled Models

- Posterior & HDI

```
az.plot_posterior(  
    idata_pooled,  
    var_names=["theta"])
```

```
plt.show()
```



Unpooled Models

- A model that estimates separate parameters for each group without sharing information across groups.
- No information sharing - a modeling structure in which data from one group do not affect the parameter estimates for other groups.

Unpooled Models

- The unpooled model estimates a separate Biotechnology startup probability *for each country*.
- There is no information sharing between countries.

```
coords = {"country": country_names}
```

```
with pm.Model(coords=coords) as unpooled_model:
```

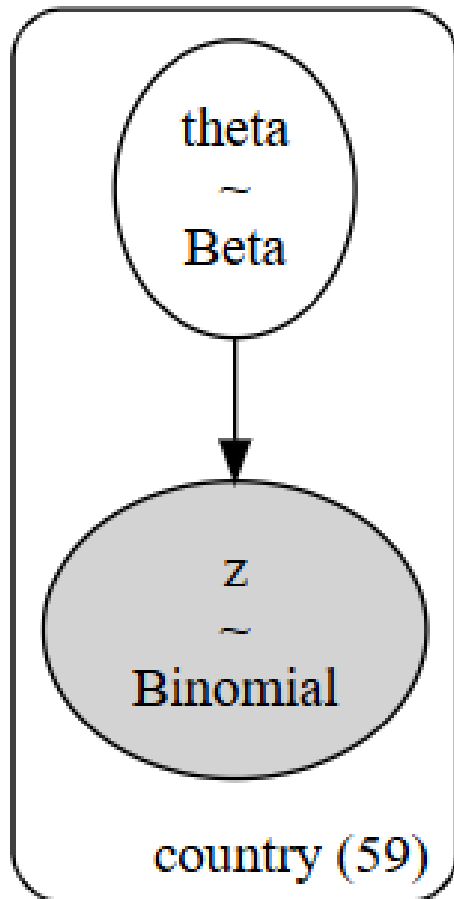
```
    # prior
```

```
    theta = pm.Beta(  
        "theta",  
        alpha=omega_alpha,  
        beta=omega_beta,  
        dims="country")
```

```
    # likelihood
```

```
    z = pm.Binomial(  
        "z",  
        n=n_startups,  
        p=theta,  
        observed=n_biotech,  
        dims="country")
```

Unpooled Models



```
coords = {"country": country_names}
```

```
with pm.Model(coords=coords) as unpooled_model:
```

```
# prior
```

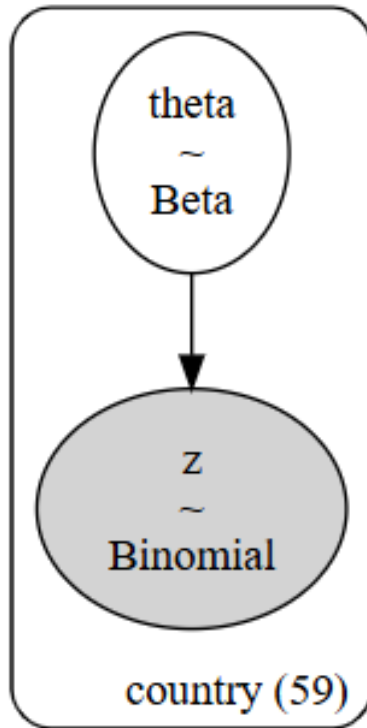
```
theta = pm.Beta(  
    "theta",  
    alpha=omega_alpha,  
    beta=omega_beta,  
    dims="country")
```

```
# likelihood
```

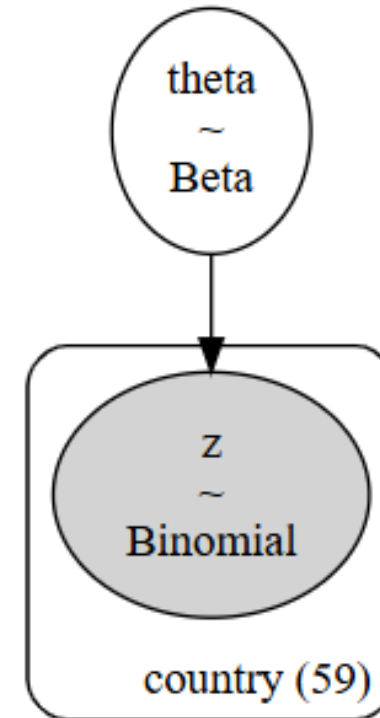
```
z = pm.Binomial(  
    "z",  
    n=n_startups,  
    p=theta,  
    observed=n_biotech,  
    dims="country")
```

Unpooled Models

Unpooled Model



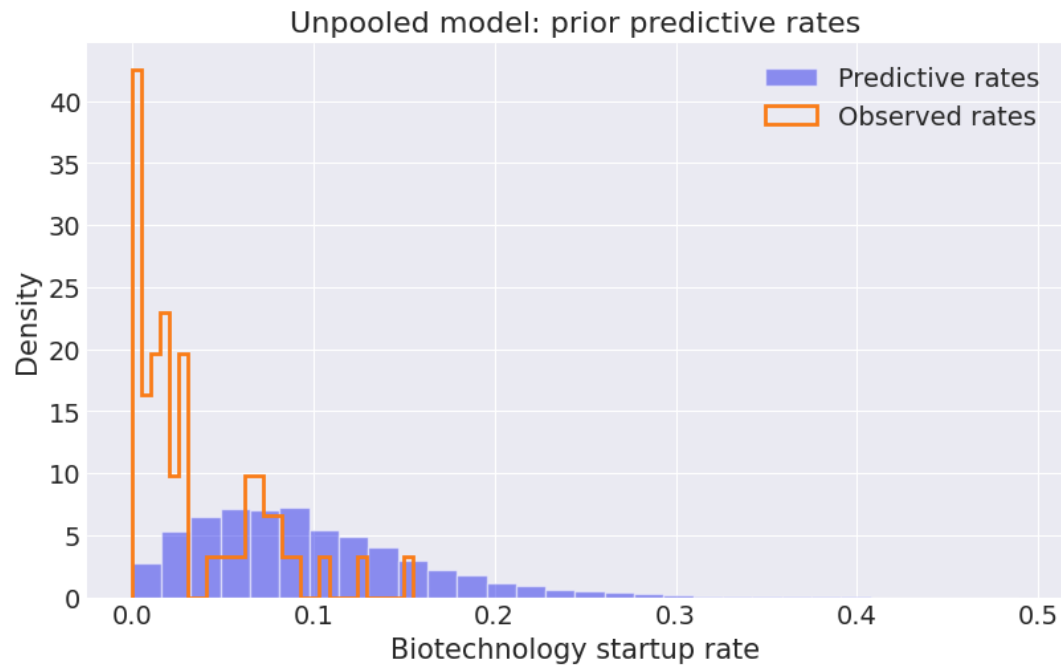
Pooled Model



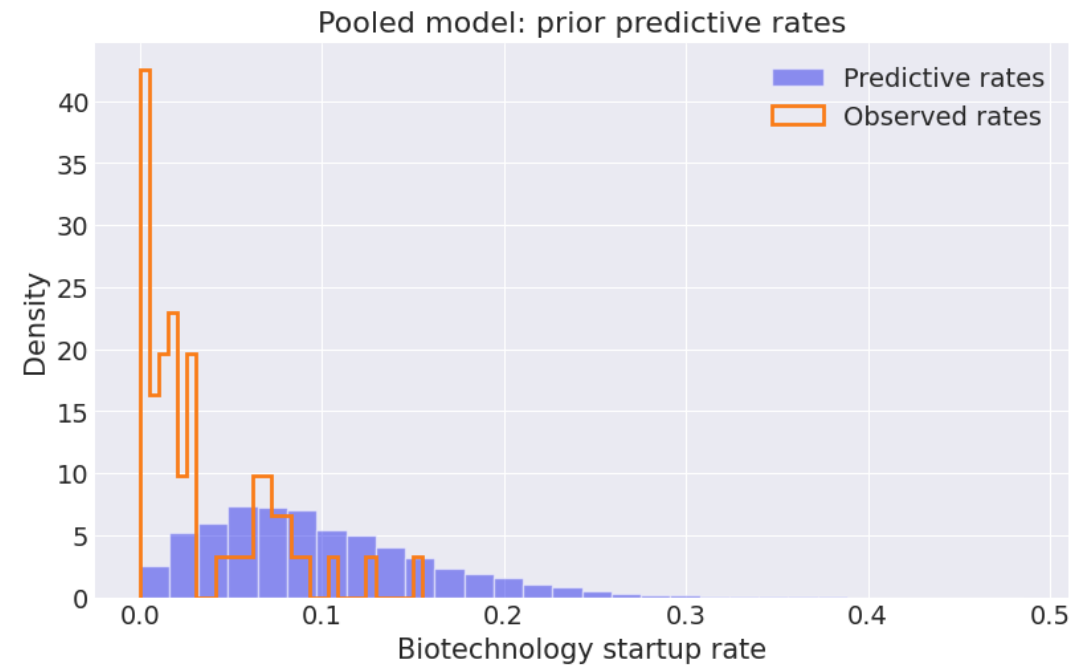
Unpooled Models

- Prior predictive

Unpooled Model



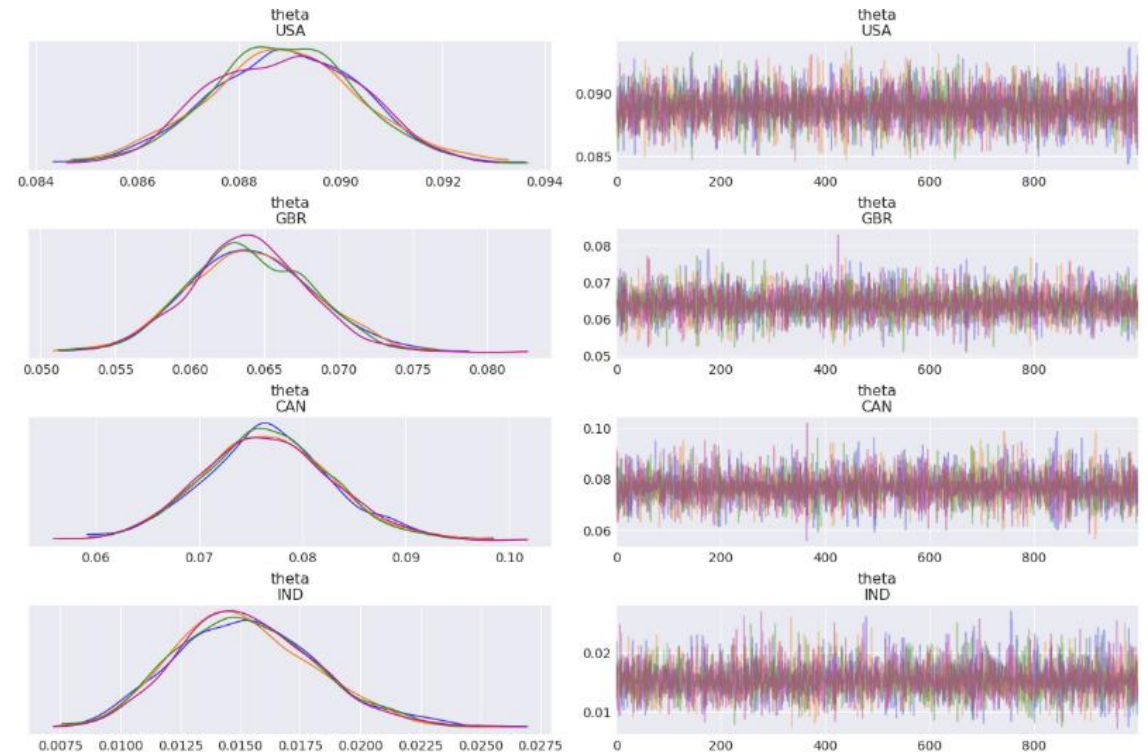
Pooled Model



Unpooled Models

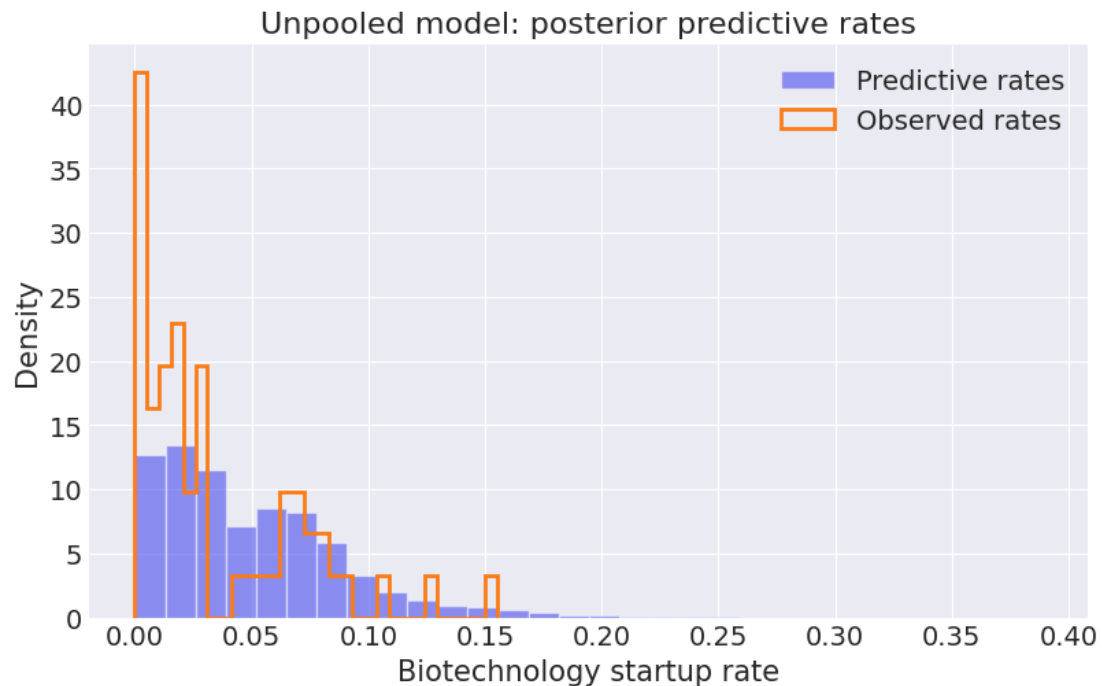
- Summary & trace plot shows the different thetas:

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta[USA]	0.0	0.000	8831.0	2958.0	1.00
theta[GBR]	0.0	0.000	6977.0	2615.0	1.00
theta[CAN]	0.0	0.000	8343.0	2655.0	1.00
theta[IND]	0.0	0.000	7320.0	2620.0	1.00

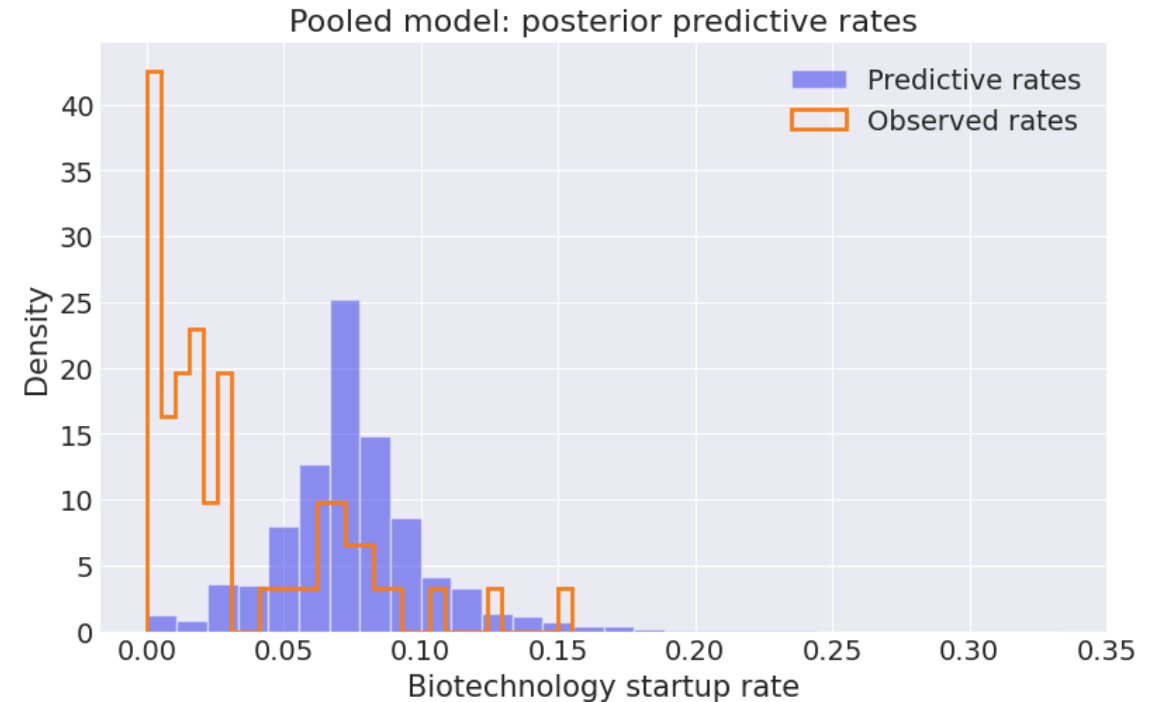


Unpooled Models

- Posterior predictive -
Unpooled Model

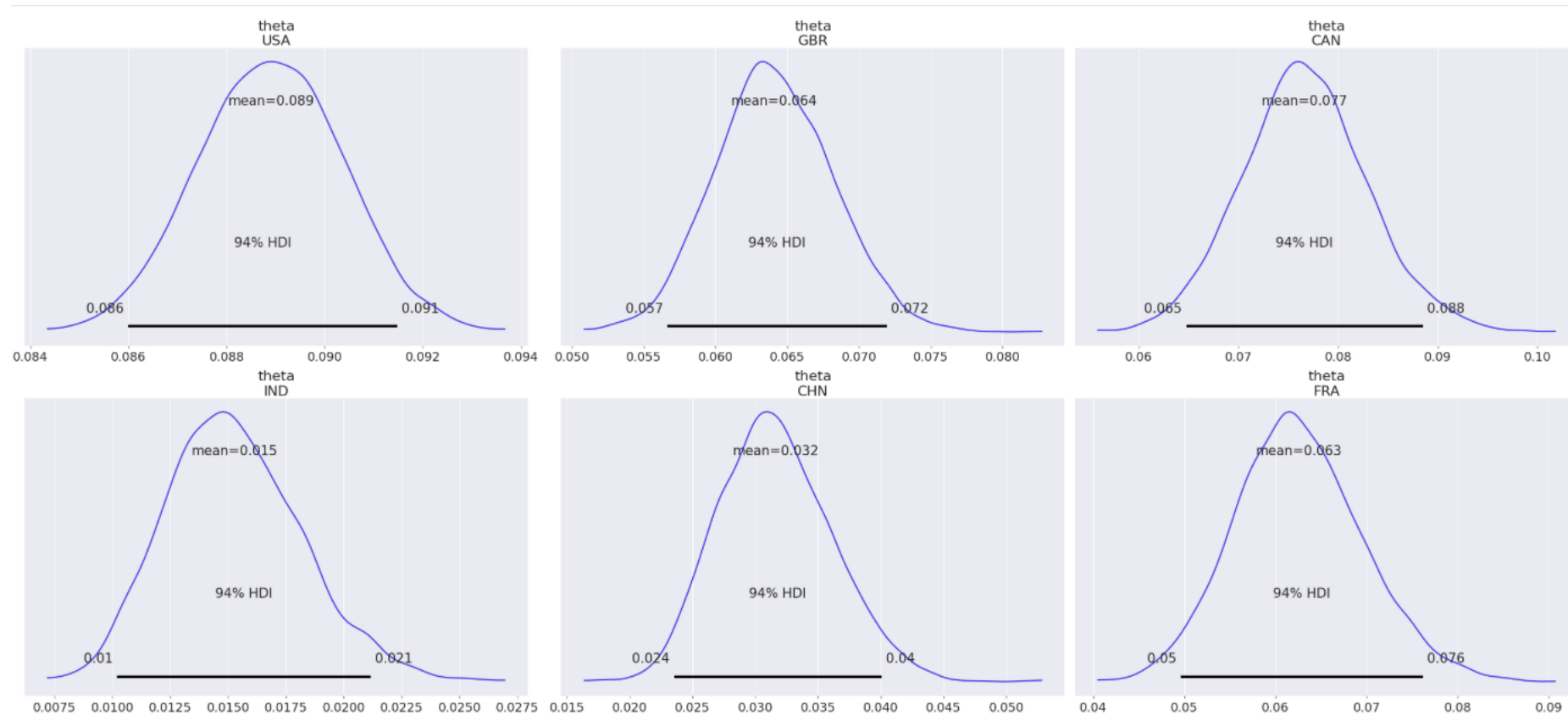


Pooled Model



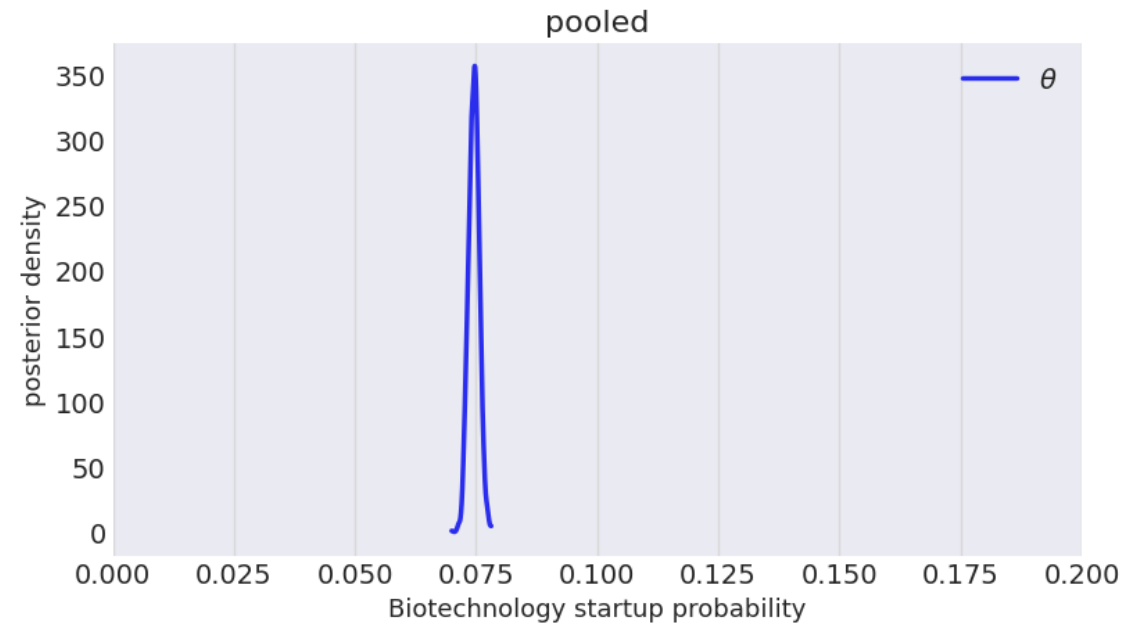
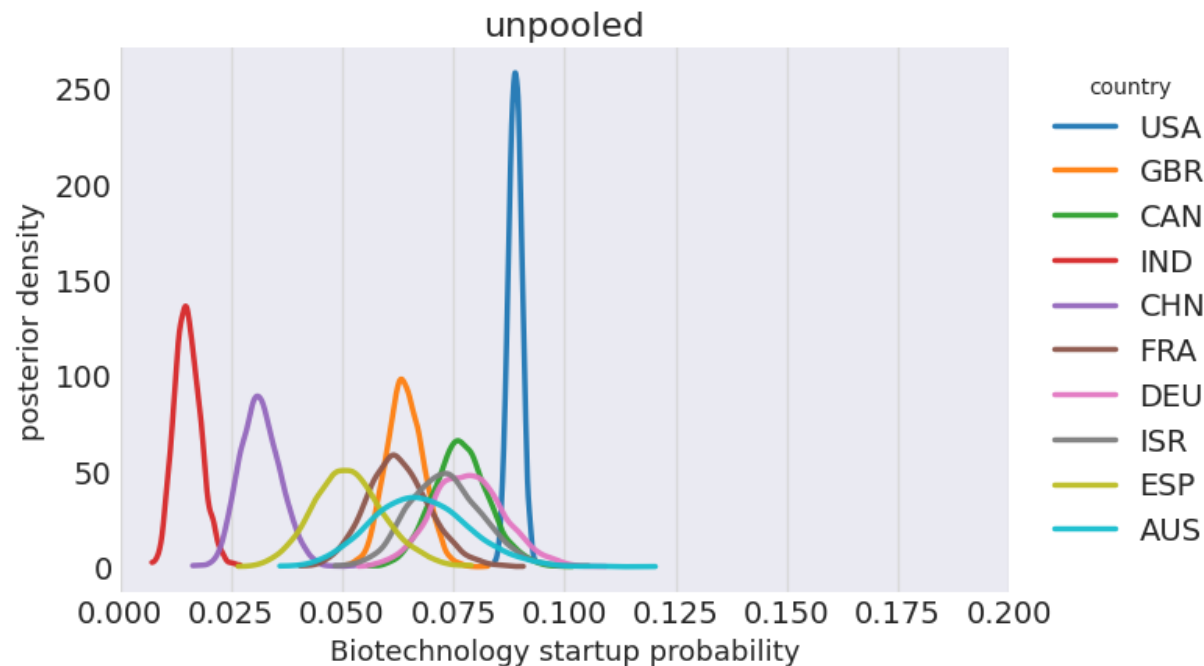
Unpooled Models

Instead of one posterior we have a different one for each country



Unpooled Models

Instead of one posterior we have a different one for each country



Hierarchical Models

Hierarchical Models

- A model in which parameters for lower-level groups are themselves modeled as coming from a higher-level distribution.
- **Hyperparameter** - a parameter of a prior distribution.
- **Hyperprior** - a prior distribution placed on a hyperparameter.
- **Information sharing** - the way group-level parameter estimates are influenced by data from other groups through shared hyperparameters.

Hierarchical Models

- In our dataset -
- The hierarchical model estimates a separate Biotechnology startup probability for each country.
- The country probabilities are connected through shared hyperparameters.
- This creates information sharing between countries.

Hierarchical Models

```
coords = {"country": country_names}

with pm.Model(coords=coords) as hierarchical_model:

    # hyperpriors
    mu_theta = pm.Beta(
        "mu_theta", alpha=omega_alpha, beta=omega_beta)

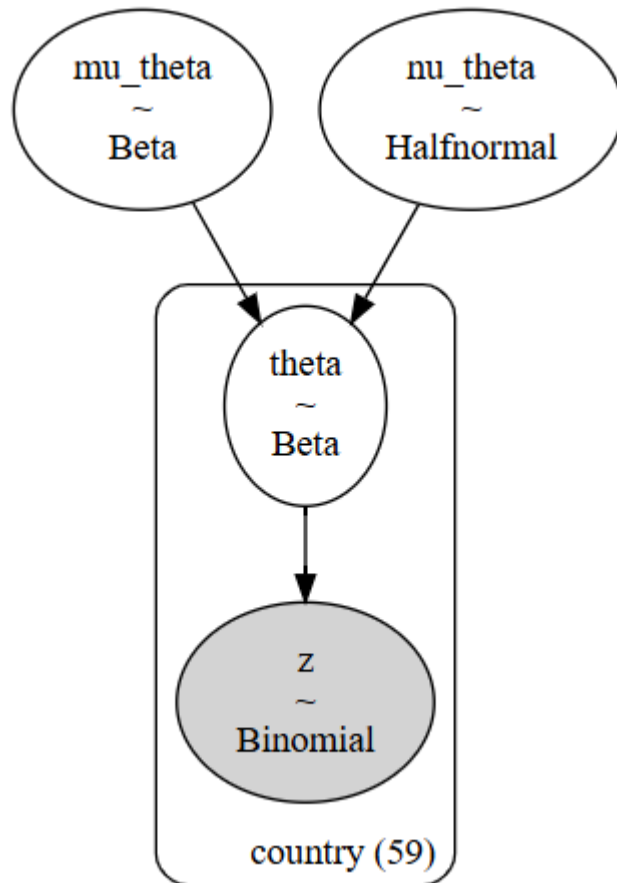
    nu_theta = pm.HalfNormal("nu_theta", sigma=10)

    # country-level prior
    theta = pm.Beta("theta", mu=mu_theta, nu=nu_theta, dims="country")

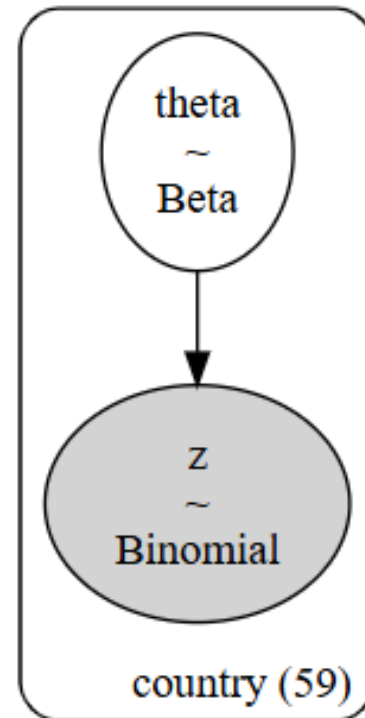
    # likelihood
    z = pm.Binomial("z", n=n_startups, p=theta, observed=n_biotech, dims="country")
```

Hierarchical Models

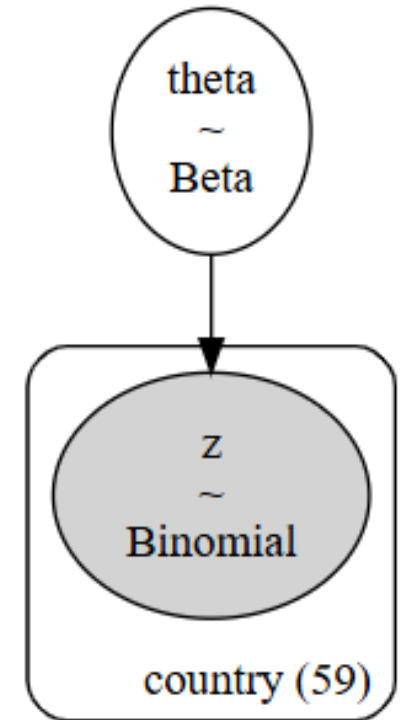
Hierarchical Model



Unpooled model

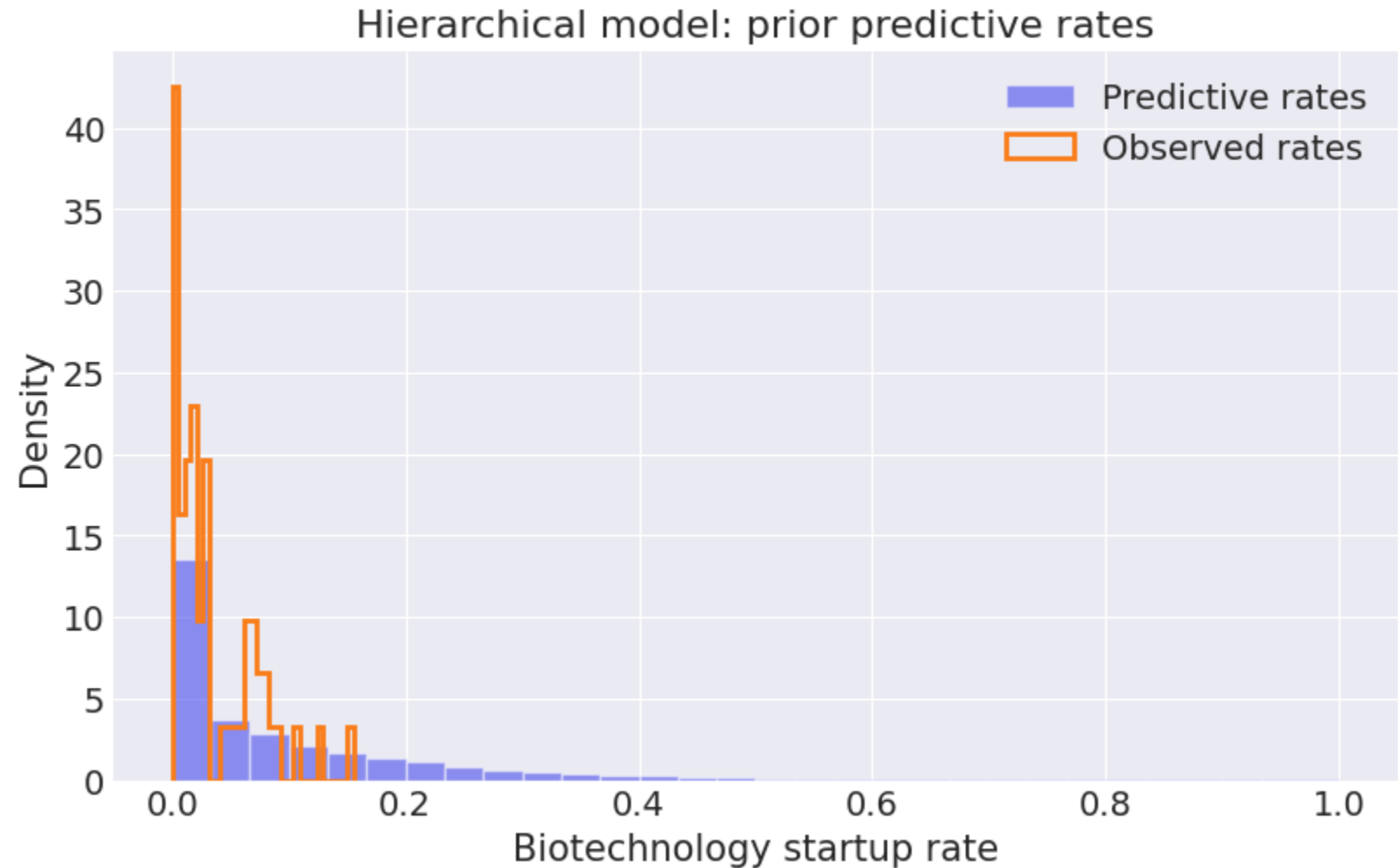


Pooled Model



Hierarchical Models

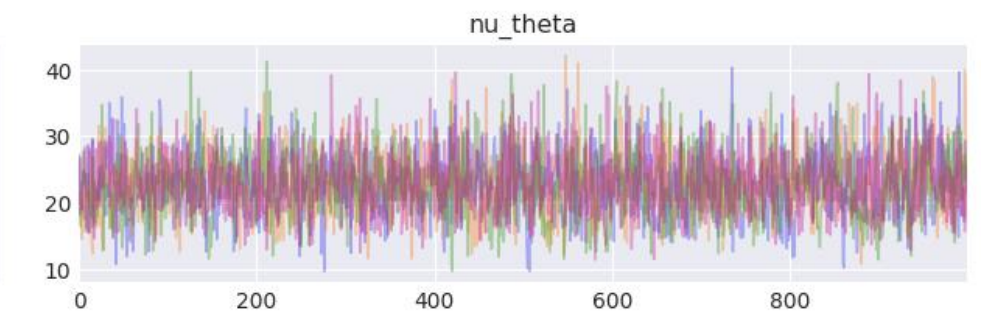
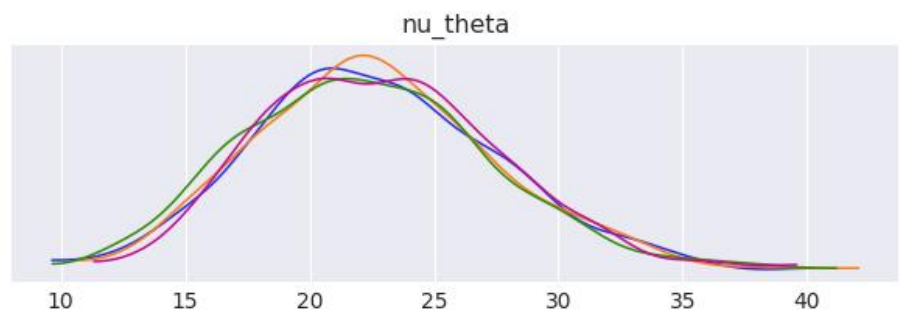
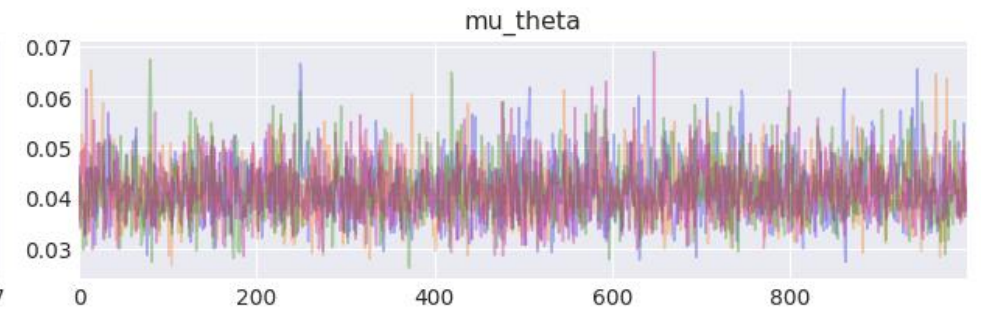
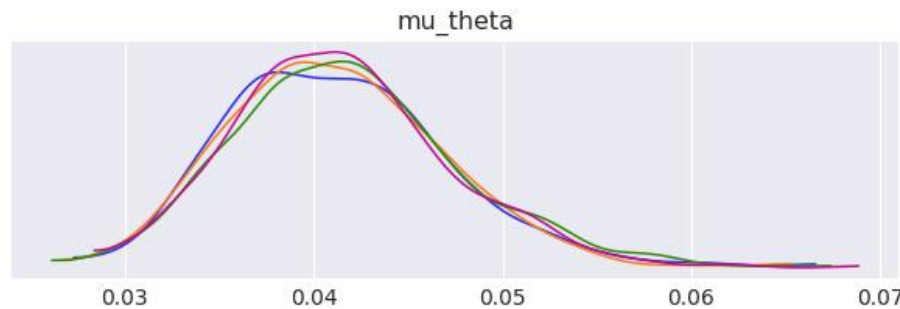
- Prior Predictive



Hierarchical Models

- Sample & diagnostics for hyperpriors

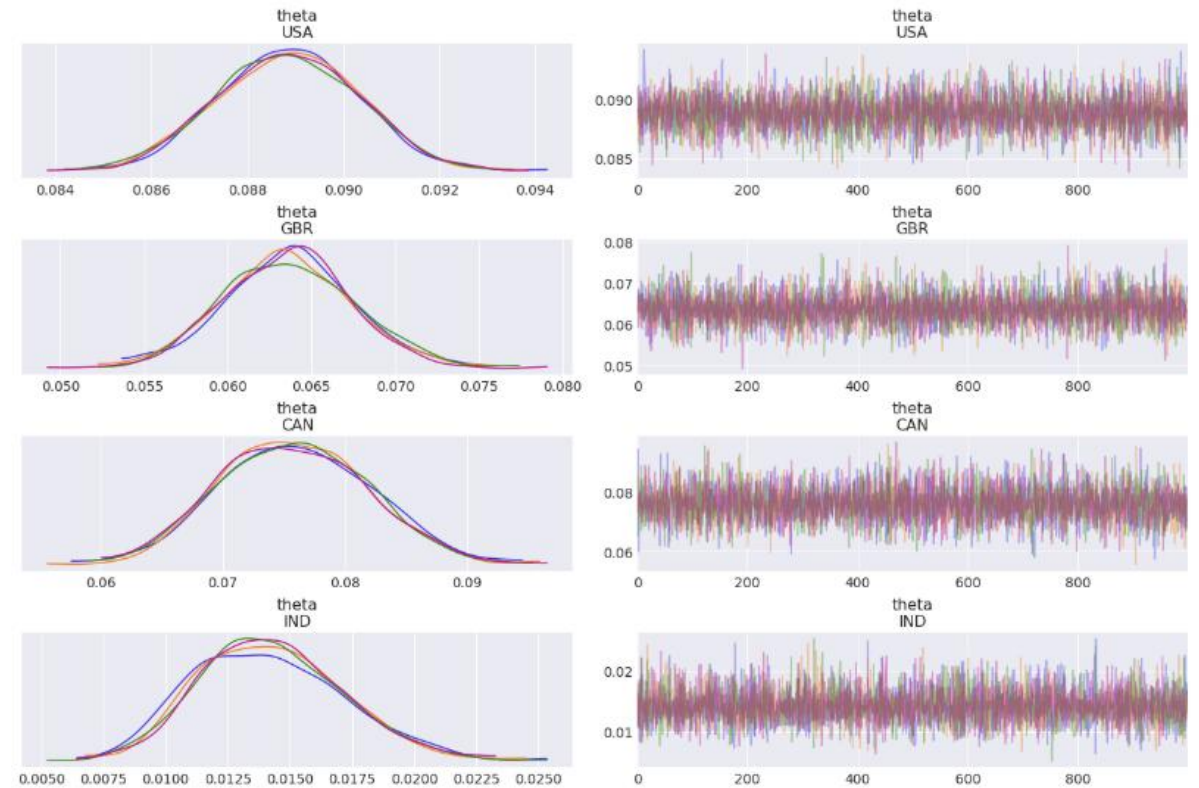
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_theta	0.000	0.00	3128.0	2919.0	1.0
nu_theta	0.109	0.07	1966.0	2475.0	1.0



Hierarchical Models

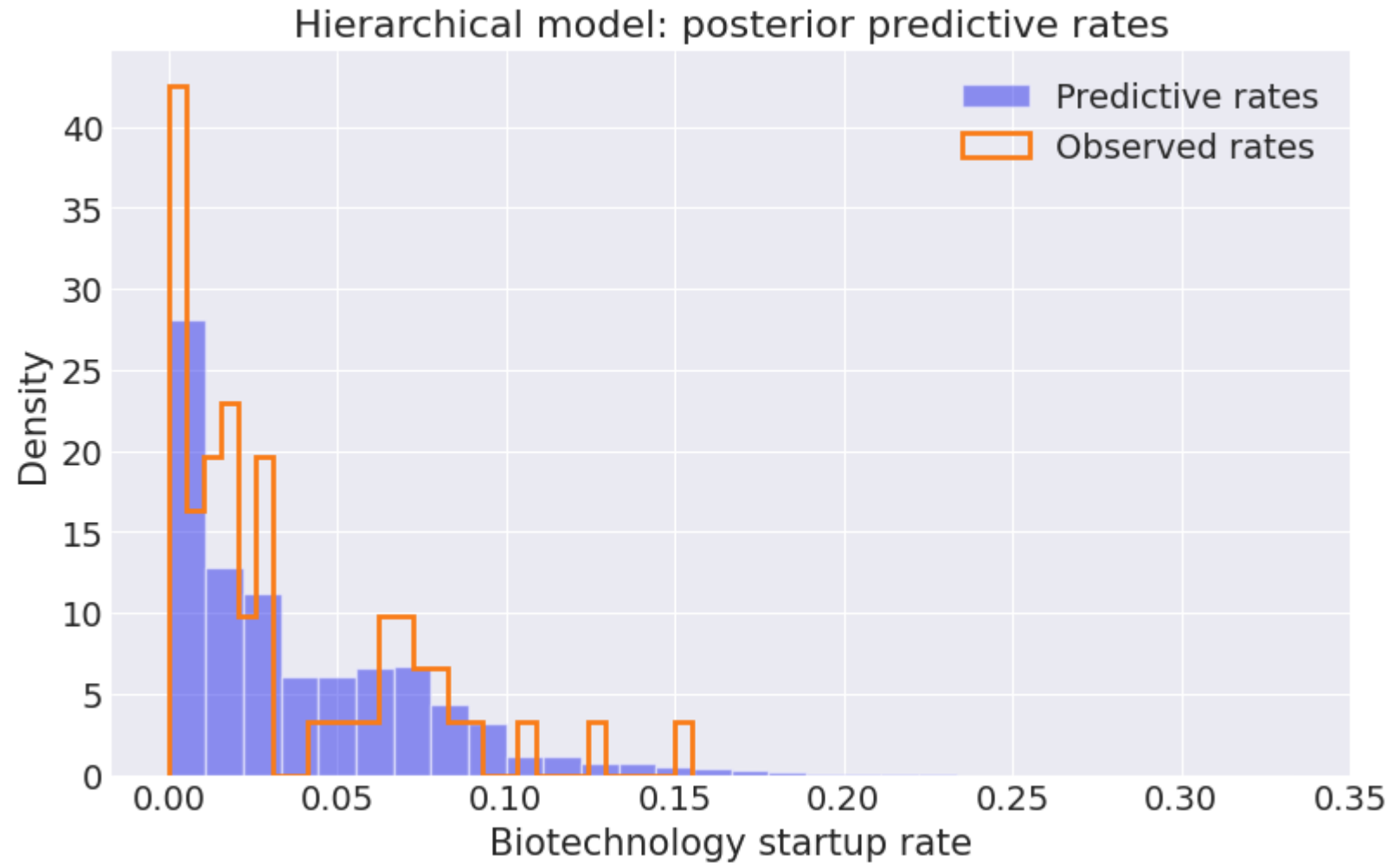
- Sample & diagnostics for country-level prior

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta[USA]	0.0	0.0	4877.0	2618.0	1.0
theta[GBR]	0.0	0.0	5487.0	2709.0	1.0
theta[CAN]	0.0	0.0	5016.0	2816.0	1.0
theta[IND]	0.0	0.0	4834.0	2895.0	1.0
theta[CHN]	0.0	0.0	5199.0	2692.0	1.0
theta[FRA]	0.0	0.0	4369.0	3067.0	1.0



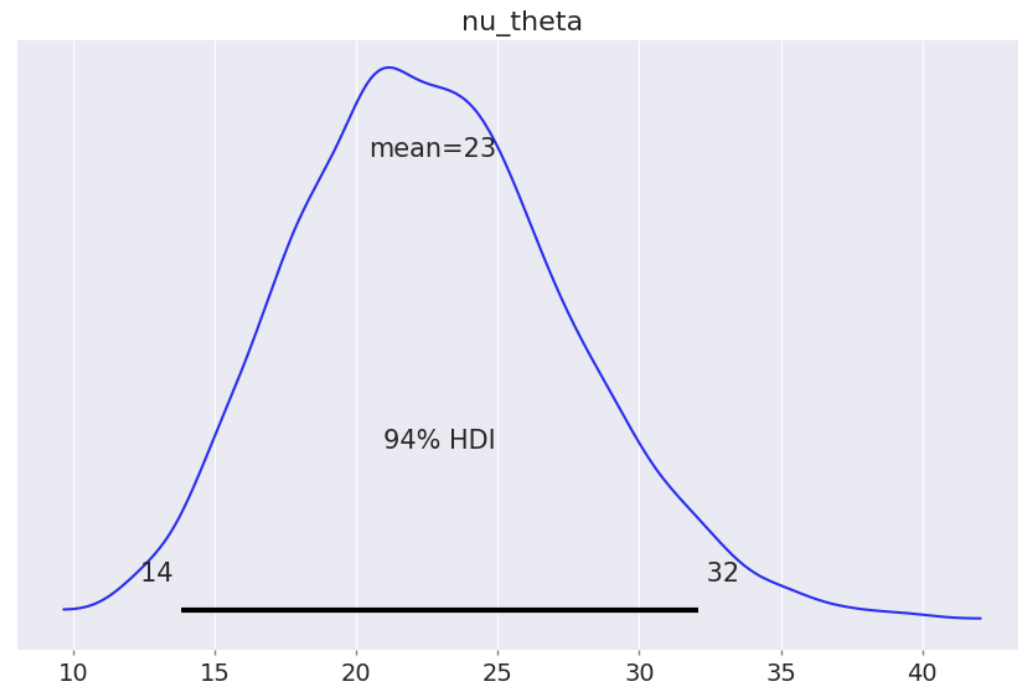
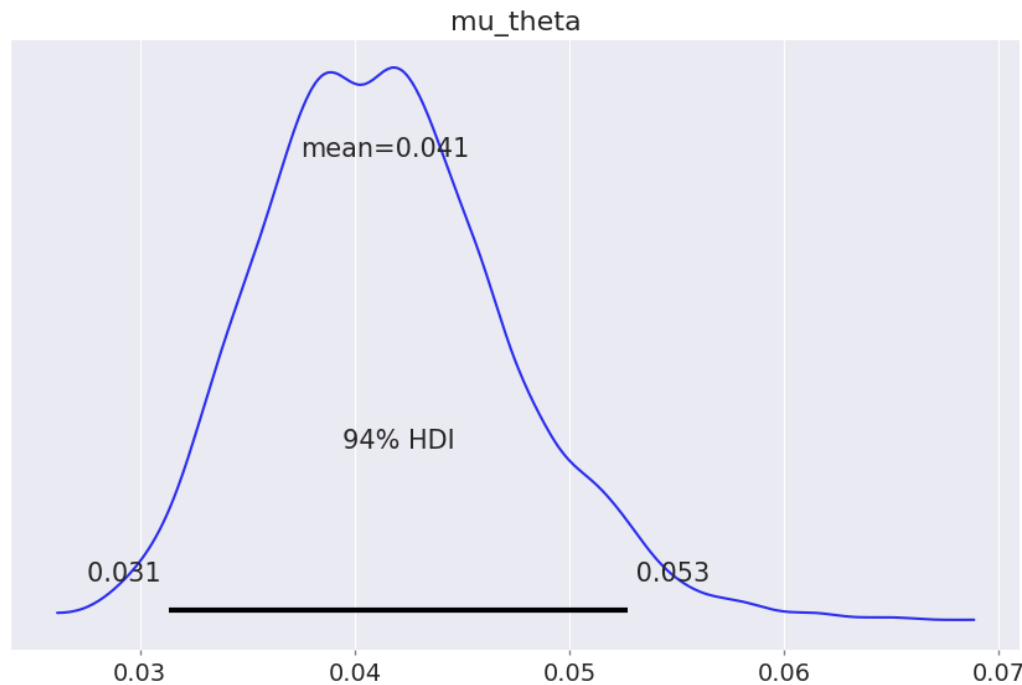
Hierarchical Models

- Posterior Predictive



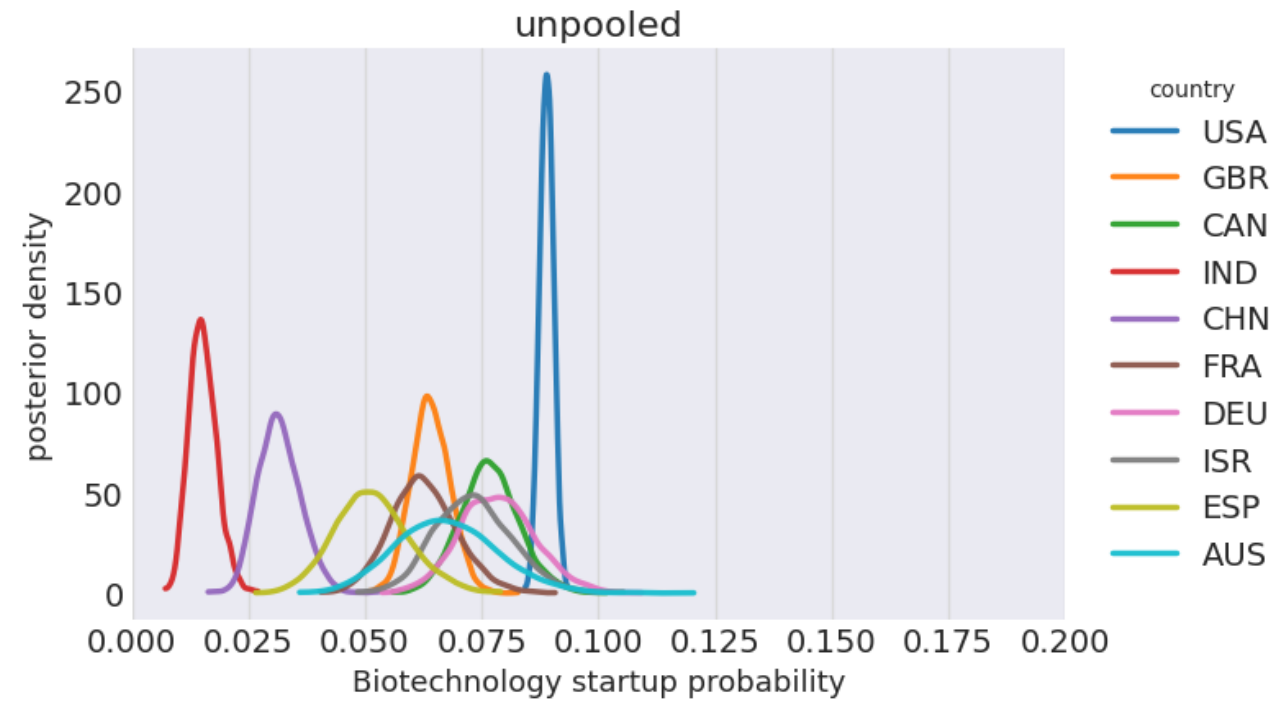
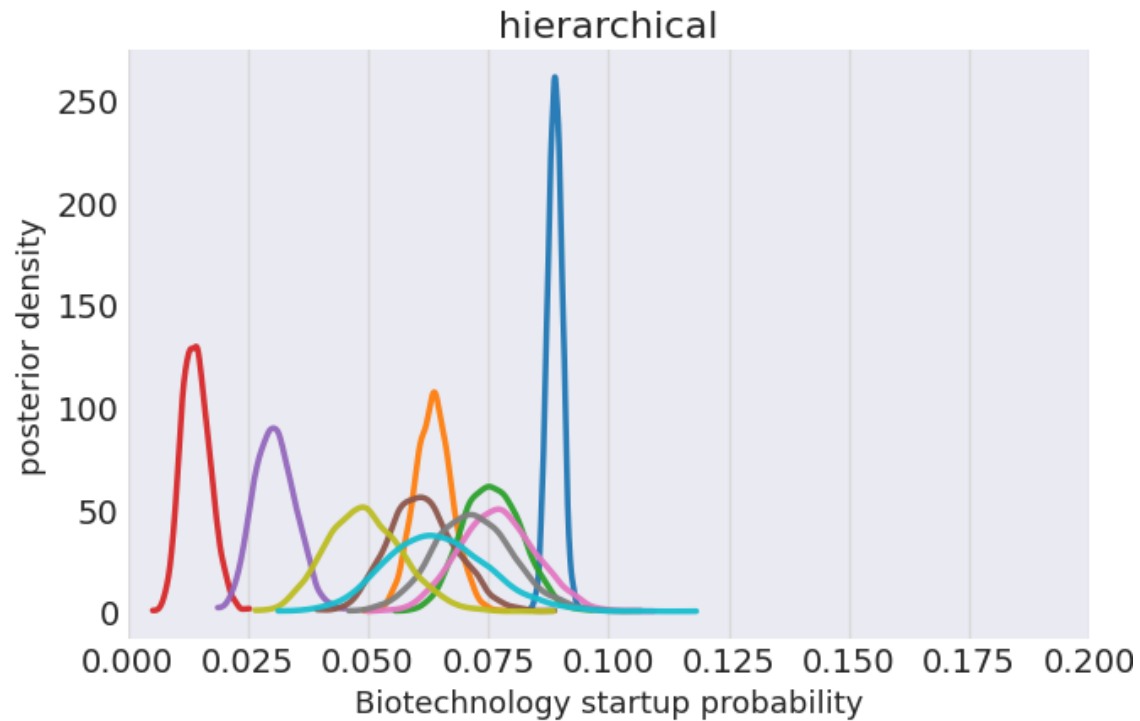
Hierarchical Models

- Posterior for the hyperparameters



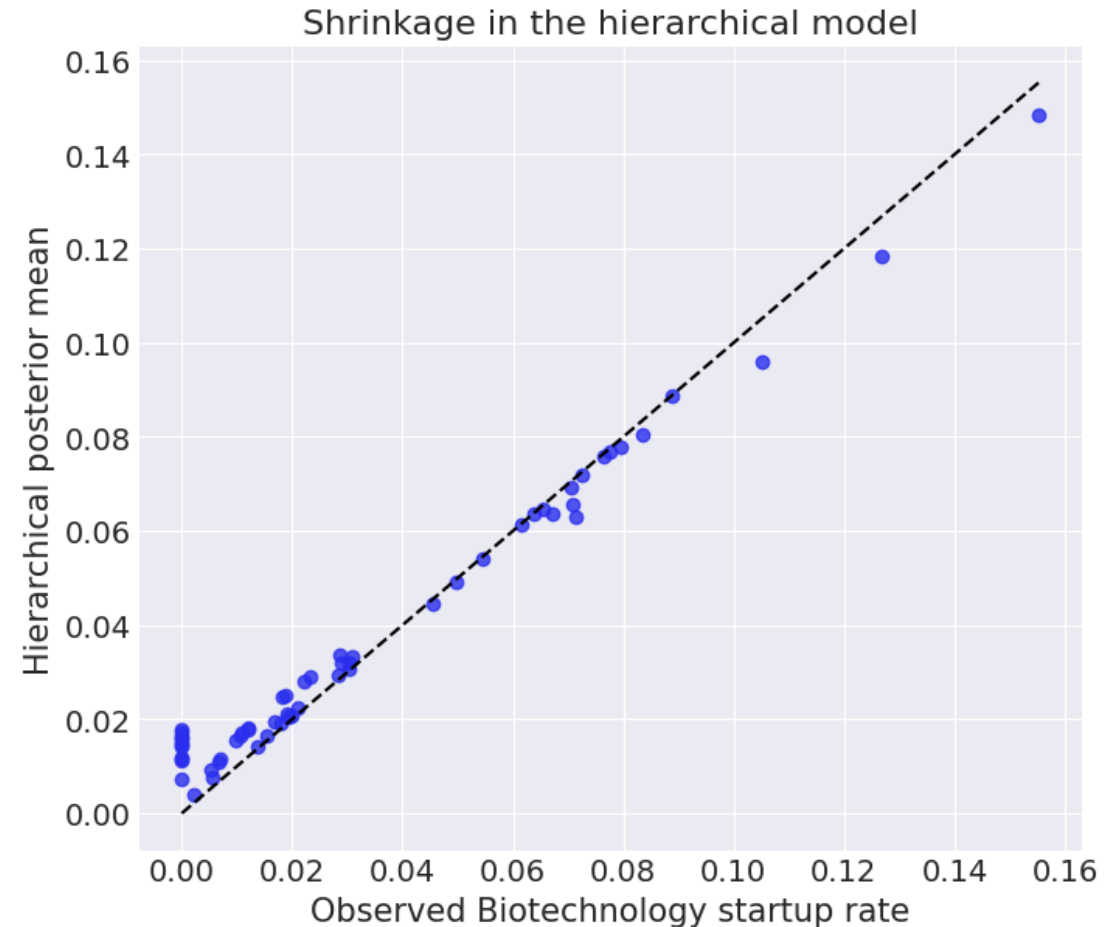
Hierarchical Models

- Posterior for theta



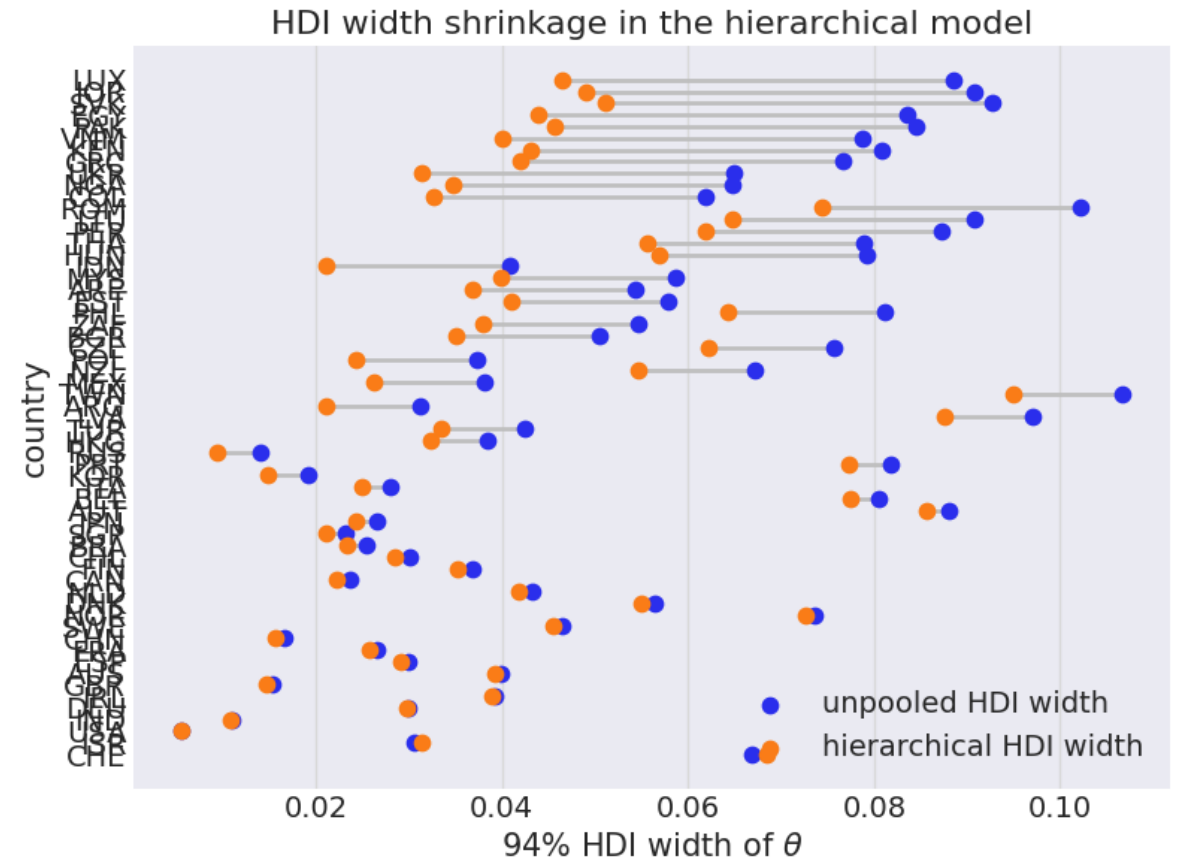
Shrinkage

- The estimates have shrunk toward the common mean due to each group informing the rest, and each group being informed by the estimation of the others.



Shrinkage

- **HDI width shrinkage** - reduction in posterior interval width, especially for groups with less data, due to hierarchical information sharing.



Multi-level Hierarchical Model

- The multi-level model adds another level of group structure.
- Each city has its own Biotechnology startup probability.
- Cities from the same country are connected through country-level hyperparameters.
- Here:
 - Cities are the lower-level groups
 - Countries are the higher-level groups

	country_code	city	n_startups	n_biotech	observed_biotech_rate
0	USA	San Francisco	3526	87	0.024674
1	USA	New York	3163	88	0.027822
2	GBR	London	1907	39	0.020451
3	USA	Austin	774	36	0.046512
4	USA	Palo Alto	769	38	0.049415

Multi-level Hierarchical Model

```
with pm.Model(coords=coords_multilevel) as multilevel_model:
    # data index:
    # for each city, this tells us which country it belongs to
    c_i = pm.Data("c_i", country_idx, dims="city")

    # hyperpriors
    mu_global = pm.Beta("mu_global", alpha=omega_alpha, beta=omega_beta)

    nu_global = pm.Gamma("nu_global", mu=10, sigma=5)

    # country-level hyperparameters
    mu_country = pm.Beta("mu_country", mu=mu_global, nu=nu_global, dims="country")

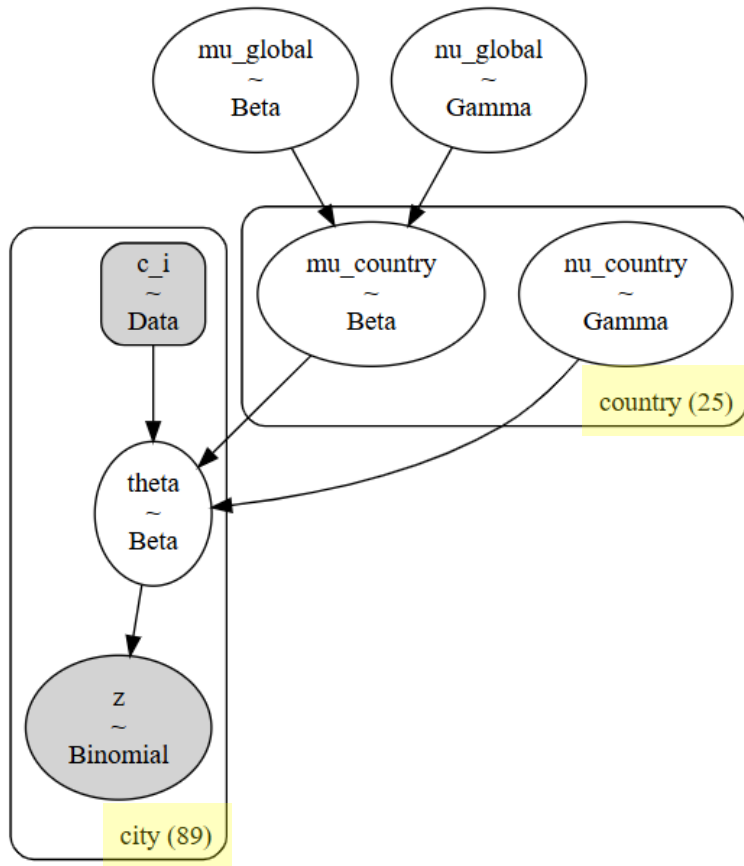
    nu_country = pm.Gamma("nu_country", mu=10, sigma=5, dims="country")

    # city-level prior
    theta = pm.Beta("theta", mu=mu_country[c_i], nu=nu_country[c_i], dims="city")

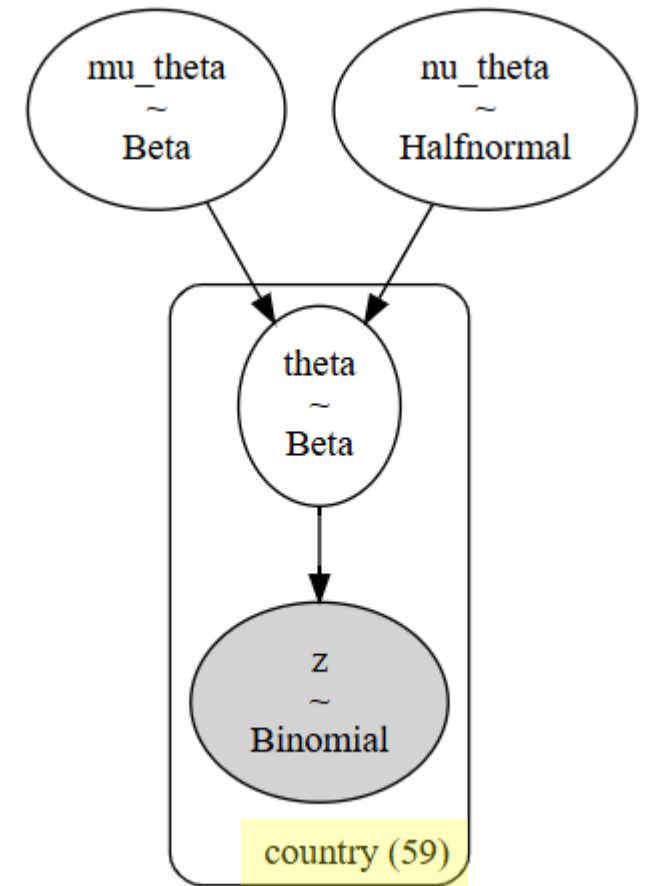
    # likelihood
    z = pm.Binomial("z", n=city_n_startups, p=theta, observed=city_n_biotech, dims="city")
```

Multi-level Hierarchical Model

Multi-level Model

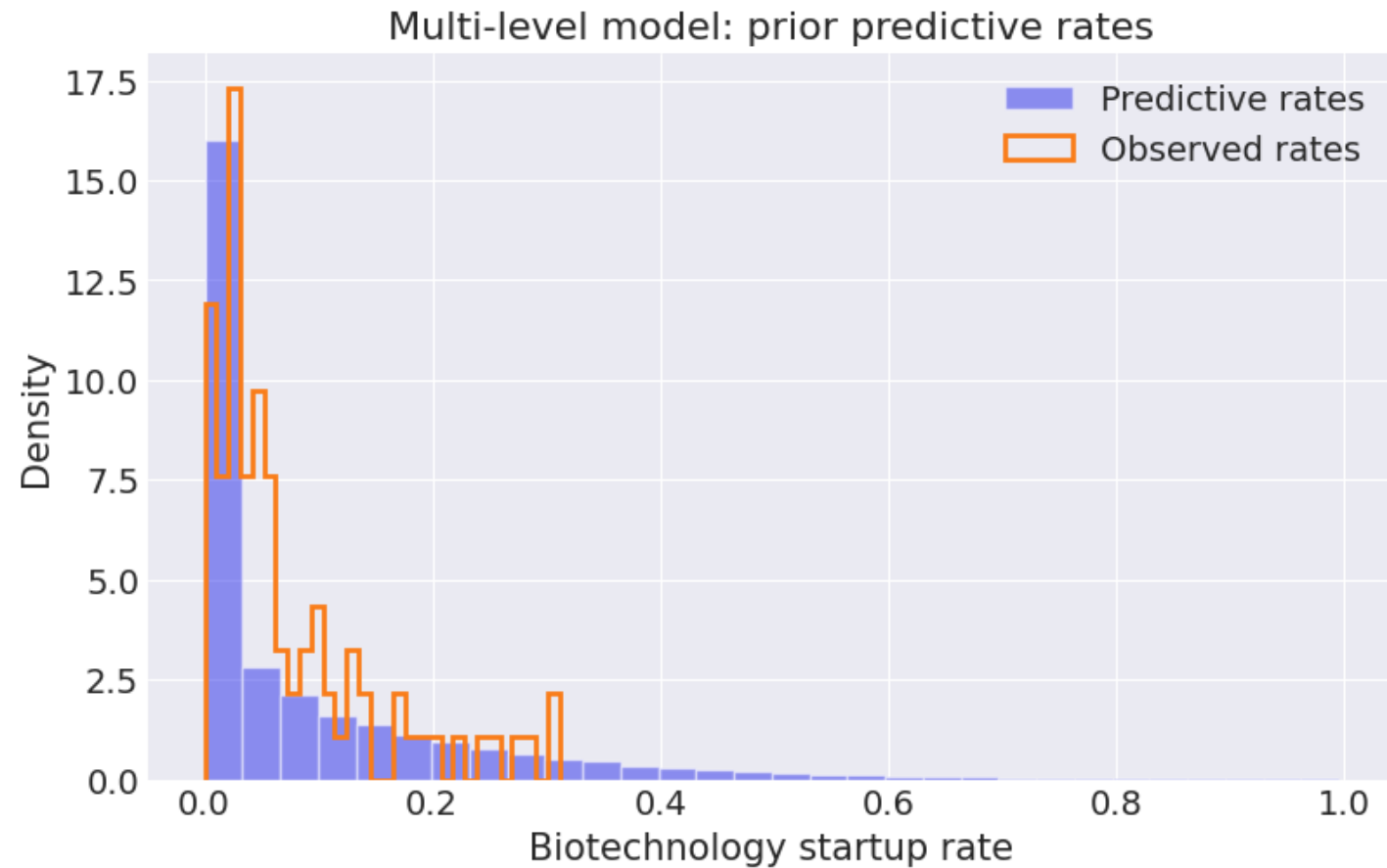


Hierarchical Model



Multi-level Hierarchical Model

- Prior predictive check



Multi-level Hierarchical Model

- Summary
 - Global hyperparameters
 - Country level hyperparameters
 - City level parameters

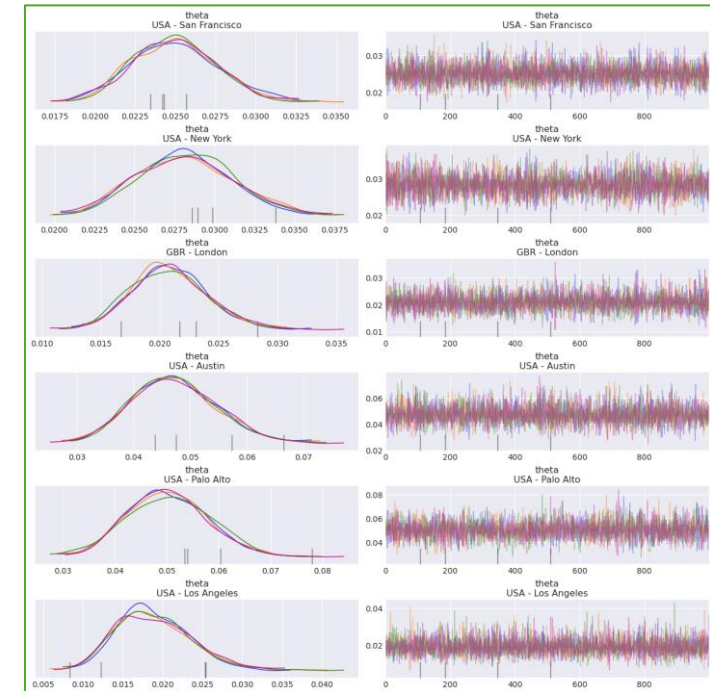
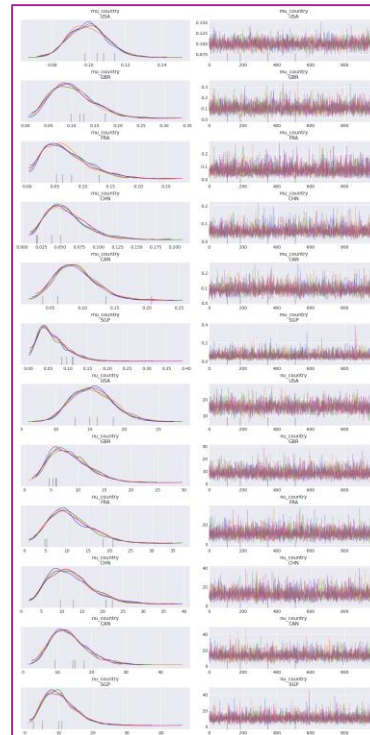
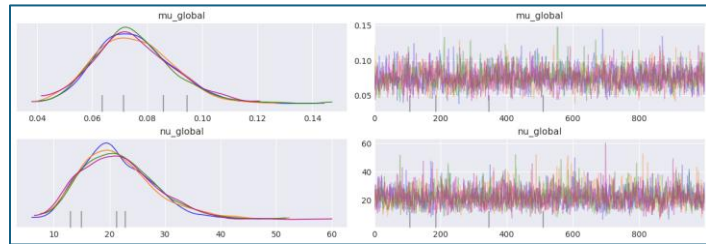
	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_global	0.000	0.000	1900.0	2256.0	1.0
nu_global	0.154	0.132	1962.0	2304.0	1.0

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_country[USA]	0.000	0.000	3200.0	2397.0	1.0
mu_country[GBR]	0.001	0.001	3590.0	2631.0	1.0
mu_country[FRA]	0.001	0.001	3558.0	2119.0	1.0
mu_country[CHN]	0.000	0.001	3112.0	2479.0	1.0
mu_country[CAN]	0.001	0.001	3480.0	2771.0	1.0
mu_country[SGP]	0.001	0.001	2752.0	1798.0	1.0
nu_country[USA]	0.055	0.050	2990.0	2520.0	1.0
nu_country[GBR]	0.058	0.063	3896.0	2222.0	1.0
nu_country[FRA]	0.071	0.093	4731.0	2575.0	1.0
nu_country[CHN]	0.086	0.095	3506.0	2262.0	1.0
nu_country[CAN]	0.080	0.095	3896.0	2491.0	1.0
nu_country[SGP]	0.070	0.115	4866.0	2657.0	1.0

	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta[USA - San Francisco]	0.0	0.0	4993.0	2525.0	1.0
theta[USA - New York]	0.0	0.0	4356.0	2543.0	1.0
theta[GBR - London]	0.0	0.0	4506.0	2497.0	1.0
theta[USA - Austin]	0.0	0.0	4329.0	2267.0	1.0
theta[USA - Palo Alto]	0.0	0.0	3123.0	1798.0	1.0
theta[USA - Los Angeles]	0.0	0.0	5262.0	2471.0	1.0

Multi-level Hierarchical Model

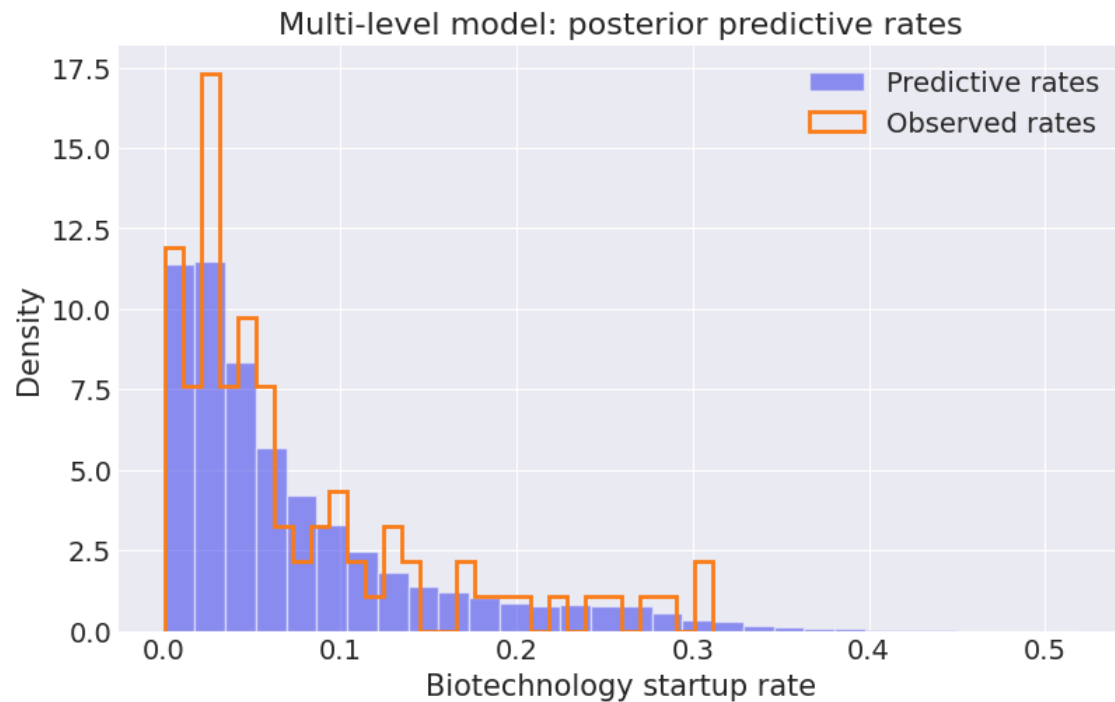
- We also have trace plots for the same parameters
 - Global hyperparameters
 - Country level hyperparameters
 - City level parameters



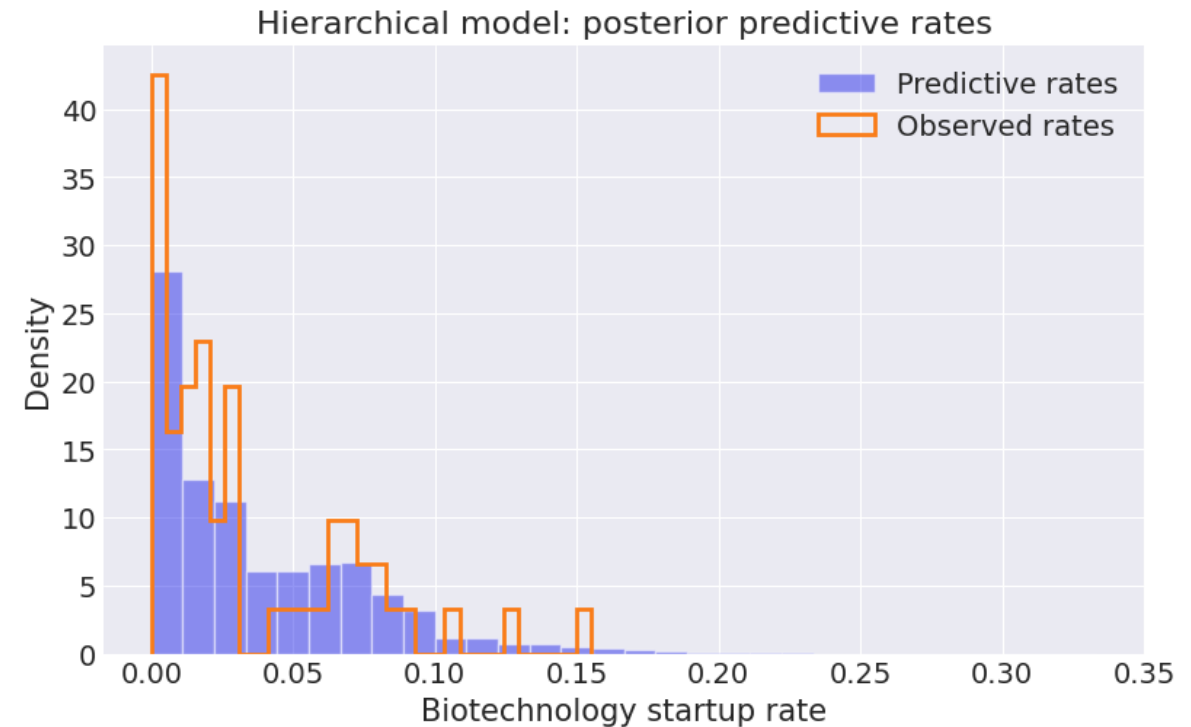
Multi-level Hierarchical Model

- Posterior Predictive check

Multi-level Hierarchical Model

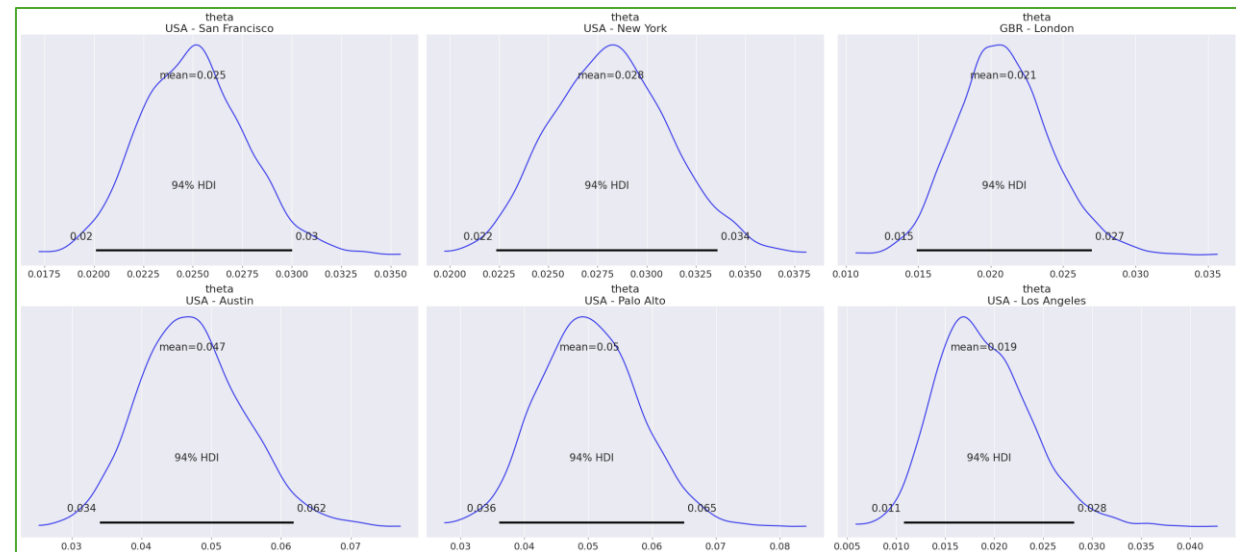
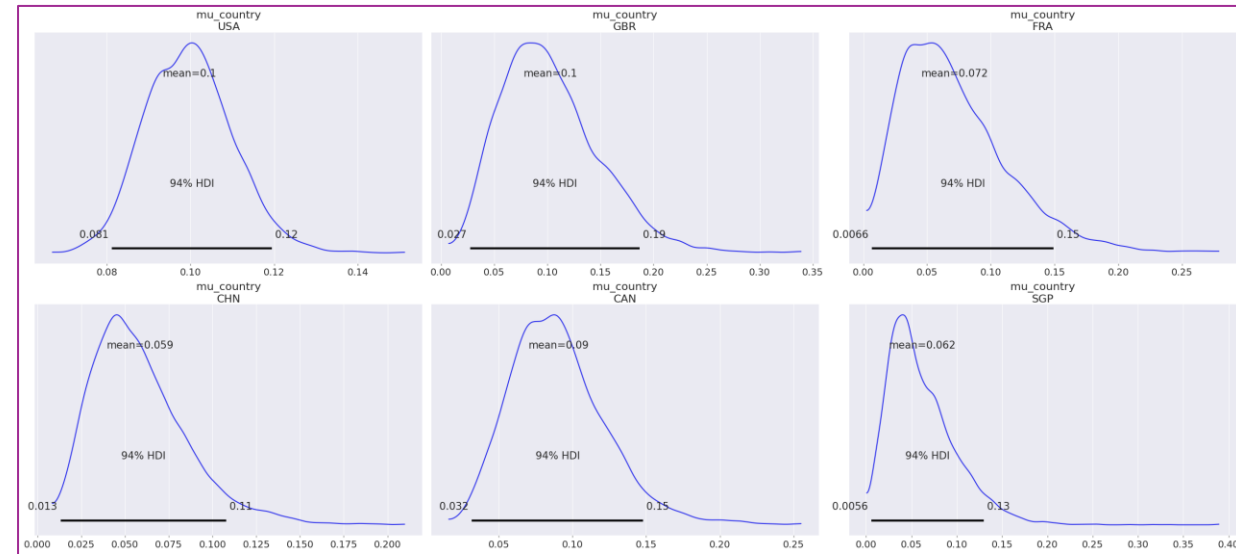
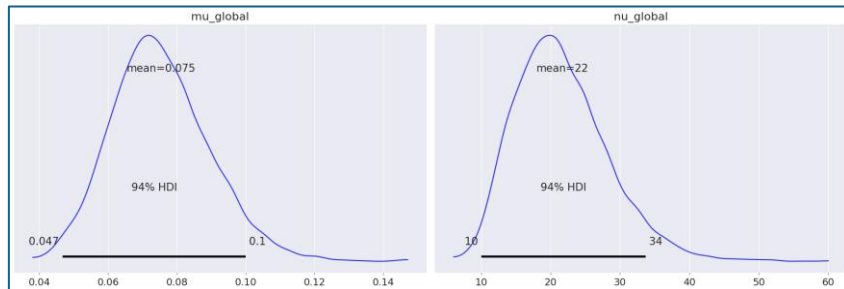


Hierarchical Model



Hierarchical Models

- Posterior
 - Global hyperparameters
 - Country level hyperparameters
 - City level parameters



Multi-level Hierarchical Model

